EXERCISE SOLUTIONS TO *THE RISING SEA: FOUNDATIONS OF ALGEBRAIC GEOMETRY*

YIMING SONG

The chapter titles, subchapter titles, exercise numberings are taken from the 2025 Princeton University Press edition of the book.

Contents

1. Just enough category theory to be dangerous

1.1. Categories and functors.

Exercise (1.1.A)**.** *Groupoids.*

(a) Make sense of the statement "a group is a groupoid with one element."

Solution. Let $\mathcal{G}$ be a one element groupoid, call the single element x. Then by definition the identity morphism is in $\hom_{\mathcal{G}}(x, x)$. We will take this to be the identity element of the group $\hom_{\mathcal{G}}(x, x)$. Any two morphisms are composable, which gives the group composition, and by definition of groupoid any morphism is invertible, yielding the group inverse.

Conversely, if we have a group G we can make a category with one object whose morphisms are the elements of G, and whose identity, composition, and inverses are given by the group structure. This yields a groupoid. □

(b) Give an example of a groupoid that is not a group.

Solution. A category with more than one object, and only the identity morphisms. □

Exercise (1.1.B)**.** *If A is an object in a category $\mathcal{C}$, show that the invertible elements of $\hom_{\mathcal{C}}(A, A)$ form a group, called the automorphism group. What are the automorphism groups in Set and Vect_k?*

Proof. This is exactly the same as part 1 above. The identity morphism serves as the identity, the inverse morphisms yield the group inverses, and the category axioms yield composition.

The automorphism groups in Set are the symmetric groups, and the automorphism groups in Vect_k are the general linear groups. ☐

Exercise (1.1.C.)**.** *Fix a field k. Let $(-)^{\vee\vee}$ be the double dual functor from the category of finite-dimensional k-vector spaces to itself. Show that $(-)^{\vee\vee}$ is naturally isomorphic to the identity functor.*

Proof. For a fixed vector space V, define the map $\alpha_V : V \to V^{\vee\vee}$ by $v \mapsto (\phi \mapsto \phi(v))$. This is injective because if $\phi \mapsto \phi(v)$ is the zero map for every ϕ, then $v = 0$, so by finite-dimensionality α_V is an isomorphism. And it remains to check that for any map of vector spaces $f : V \to W$ and $v \in V$, that

$$\begin{aligned}\alpha_W(f(v)) &= (\phi \mapsto \phi(f(v)))\\ &= (\psi \mapsto \psi(v)) \circ f^\vee\\ &= f^{\vee\vee}(\psi \mapsto \psi(v))\\ &= f^{\vee\vee}(\alpha_V(v))\end{aligned}$$

which completes the proof. ☐

Exercise (1.1.D)**.** *Let $\mathcal{V}$ be the category whose objects are the k-vector spaces k^n for $n \geq 0$, and whose morphisms are linear transformations. There is a functor F from $\mathcal{V}$ to the category of f.d. k-vector spaces sending each vector space to itself. Show that F is an equivalence of categories.*

Proof. Following Vakil's remark about ignoring set-theoretic conditions, choose a basis for each finite dimensional k-vector space. For any vector space of the form k^n, choose the standard basis $\{e_i\}$.

Then define a functor G from f.d. k-vector spaces to $\mathcal{V}$ by sending an n-dimensional k-vector space to k^n. For a morphism ϕ from V to an m-dimensional vector space W, let $\{v_1, ..., v_n\}$ be the chosen basis for V and $\{w_1, ..., w_m\}$ the chosen basis for W. Then define $G\phi$ to be the matrix whose (i, j)th entry is the coefficient of w_j in $\phi(v_i)$.

Define a natural transformation α from FG to the identity functor where $\alpha_V : FGV \to V$ sends $FGV = k^n$ to V by mapping the standard basis vector e_i of FGV to the chosen ith basis vector v_i of V. Then α_V is an isomorphism. For naturality, we have for a map $\phi : V \to W$ that

$$\begin{aligned}\phi(\alpha_V(e_i)) &= \phi(v_i)\\ &= \sum_j (G\phi)_{i,j} w_j\\ &= \alpha_W(\sum_j (G\phi)_{i,j} e_j)\\ &= \alpha_W(FG\phi)(e_i).\end{aligned}$$

In the other direction, GF is the identity functor, which completes the proof. ☐

1.2. **Universal properties determine an object up to unique isomorphism.**

Exercise (1.2.A)**.** *Show that any two initial objects are uniquely isomorphic, and any two final objects are uniquely isomorphic.*

Proof. Let i_1, i_2 be initial. Then there is exactly one map $f_{12} : i_1 \to i_2$ and likewise $f_{21} : i_2 \to i_1$. Since there is also exactly one map from i_1 to itself, the identity, so because $f_{21} \circ f_{12}$ is one such map, it must be the identity. Likewise $f_{12} \circ f_{21}$ is the identity, which yields the unique isomorphisms. The argument is exactly the same for final objects. ☐

Exercise (1.2.B). *What are the initial and final objects (if they exist) in* Ring, Set, *and* Top, *or the category of subsets of a set under inclusion, or the category of open subsets of a topological space under inclusion?*

Solution. In Ring, the initial object is the integers, since 1 mapping to 1 dictates where the remaining integers go; the terminal object is the zero ring, since everything must map to zero. In Set, the initial object is the empty set; the final object is the singleton. In Top, the initial object is the empty space; the final object is the singleton. In the category of subsets or open sets, the initial object is the empty set/space; the terminal object is the entire set/space. □

Exercise (1.2.C). *Let A be a commutative ring and S a multiplicative subset of A. Show that $f : A \mapsto S^{-1}A$ via $a \mapsto a/1$ is injective if and only if S contains no zerodivisors.*

Proof. First suppose f is not injective, i.e. there is a nonzero element $a \in A$ such that $a/1 = 0/1$. So there exists some $s \in S$ such that $s(a1 - 01) = sa = 0$. Then s is a zerodivisor.

Conversely, suppose S contains a zerodivisor s, with $st = 0$ for t nonzero. Then $0 = st = s(t1 - 01)$, $t/1 = 0/1 = 0$, so $f(t) = t/1 = 0/1$ and f fails to be injective. □

Exercise (1.2.D). *Verify that $\iota : A \to S^{-1}A$ is initial among A-algebras B where every element of S is sent to an invertible element in B, i.e. any map $\phi : A \to B$ sending all elements of S to invertible elements in B factors through $S^{-1}A$.*

Proof. Define $\tilde{\phi} : S^{-1}A \to B$ via $\tilde{\phi}(a/s) := \phi(a)\phi(s)^{-1}$. This makes sense since we have assumed that $\phi(s)$ is invertible. Let $a, b \in A$ and $s, t \in S$. Then

$$\tilde{\phi}(ab/st) = \phi(ab)\phi(st)^{-1} = \phi(a)\phi(b)\phi(s)^{-1}\phi(t)^{-1} = \tilde{\phi}(a/s)\tilde{\phi}(b/t)$$

and

$$\begin{aligned}
\tilde{\phi}(a/s + b/t) &= \tilde{\phi}((at + sb)/st) \\
&= \phi(at + sb)\phi(st)^{-1} \\
&= \phi(a)\phi(t)/\phi(s)\phi(t) + \phi(s)\phi(b)/\phi(s)\phi(t) \\
&= \tilde{\phi}(a/s) + \tilde{\phi}(b/t).
\end{aligned}$$

so $\tilde{\phi}$ is a ring homomorphism. And $\tilde{\phi}(\iota(a)) = \tilde{\phi}(a/1) = \phi(a)/\phi(1) = \phi(a)$ as desired. □

Exercise (1.2.E). *The localization of an A-module M defined via the universal property, actually exists.*

Proof. Following the hint, define $S^{-1}M$ to have elements of the form m/s where $m \in M$ and $s \in S$, subject to the equivalence relation where $m/s \sim m'/t$ if there exists an element $s' \in S$ such that $s'(ms - m't) = 0$, with the standard operation and $S^{-1}A$-module operation. Define $\iota : M \to S^{-1}M$ via $m \mapsto m/1$.

Let $\phi : M \to N$ be an A-module map such that multiplication by any element of s is an isomorphism $N \to N$. Then we can define a map $\tilde{\phi} : S^{-1}M \to N$ via $m/s \mapsto s^{-1}\phi(m)$, which makes sense taking s^{-1} to be the inverse to multiplication by s. This is an A-module map and $\tilde{\phi}(\iota(m)) = \tilde{\phi}(a/1) = 1^{-1}\phi(a) = \phi(a)$. □

Exercise (1.2.F). *Properties of localization.*

(a) *Localization commutes with finite products, i.e. for A-modules $M_1, \ldots, M_n$ provide an A-module isomorphism $S^{-1}(M_1 \times \cdots M_n) \to S^{-1}M_1 \times \cdots \times S^{-1}M_n$.*

Proof. Define our map via $(m_1, \ldots, m_n)/s \mapsto (m_1/s, \ldots, m_n/s)$. For surjectivity, given an element $(m_1/s_1, \ldots, m_n/s_n)$, the element

$$(s_2 \cdots s_n m_1, \ldots, s_1 \cdots s_{n-1} m_n)/s_1 \cdots s_n$$

will map to it. For injectivity, suppose $(m_1/s, \ldots, m_n/s)$ is zero, i.e. there exist elements $t_i \in S$ such that $t_i m_i = 0$ for all i. Then $t_1 \cdots t_n (m_1, \ldots, m_n) = 0$, so $(m_1, \ldots, m_n)/s = 0$. □

(b) *Localization commutes with arbitrary direct sums.*

Proof. Let Λ be an indexing set. Define a map $S^{-1}(\bigoplus_{\lambda \in \Lambda} M_\lambda) \to \bigoplus_{\lambda \in \Lambda} S^{-1} M_\lambda$ via

$$(m_\lambda)_{\lambda \in \Lambda}/s \mapsto (m_\lambda/s)_{\lambda \in \Lambda}.$$

For surjectivity, given an element $(m_\lambda/s_\lambda)_{\lambda \in \Lambda}$, we know that only finitely many indices are nonzero, say $\lambda_1, \ldots, \lambda_k$. Then let $s = \lambda_1 \cdots \lambda_k$, and consider the element $(k_\lambda m_\lambda)_{\lambda \in \Lambda}$ where $k_{\lambda_i} = s/s_i$ for the nonzero indices and zero otherwise. Then it maps to what we want. For injectivity, suppose $(m_\lambda/s)_{\lambda \in \Lambda}$ is zero, i.e. there exist elements t_λ such that $t_\lambda m_\lambda = 0$ for all λ. Again only finitely many of the m_λ are nonzero, so take the corresponding t_λ and let their product be t. Then $t(m_\lambda)_{\lambda \in \Lambda} = 0$ and we are done. □

(c) *Localization does not commute with infinite products, i.e. the same map defined as above is not always an isomorphism for infinite products.*

Proof. We know that $\mathbf{Z}$ localized at $\mathbf{Z} \setminus \{0\}$ is $\mathbf{Q}$, so that $\prod_{\mathbf{N}} \mathbf{Q}$ contains for example the element $x := (1, 1/2, 1/3, \ldots)$. On the other hand, suppose there were an element in $(\mathbf{Z} \setminus \{0\})^{-1}(\mathbf{Z}^{\mathbf{N}})$ mapping to x, i.e. some element $(a_1, a_2, \ldots)/s$ where $a_i \in \mathbf{Z}$ and $s \in \mathbf{Z} \setminus \{0\}$ such that $a_n/s = 1/n$ for all $n \in \mathbf{N}$. Then $na_n = s$ for all $n \in \mathbf{N}$, i.e. s is an integer divisible by all natural numbers, a contradiction. □

Exercise (1.2.G)**.** *Show that* $\mathbf{Z}/10 \otimes \mathbf{Z}/12 \cong \mathbf{Z}/2$.

Proof. We define a map $\tilde{f} : \mathbf{Z}/10 \times \mathbf{Z}/12 \to \mathbf{Z}/2$ via $([a], [b]) \mapsto [ab]$. This is clearly bilinear. If $a = a' + 10n$ and $b = b' + 12m$, so $[ab] = [a'b' + 12ma' + 10nb' + 120nm] = [a'b']$ in $\mathbf{Z}/2$, so this map is well defined. By the universal property of tensor products, f factors through the tensor product to yield a map $f : \mathbf{Z}/10 \otimes \mathbf{Z}/12 \to \mathbf{Z}/2$ such that $[a] \otimes [b] \mapsto [ab]$.

Consider the map $g : \mathbf{Z}/2 \to \mathbf{Z}/10 \otimes \mathbf{Z}/12$ given by $[a] \mapsto [a] \otimes [1]$. This is again bilinear, and if $a = a' + 2n$, then $[a] \otimes [1] = [a' + 2n] \otimes [1] = [a'] \otimes [1] + [12n - 10n] \otimes [1] = [a'] \otimes [1] + [1] \otimes [12n] - [10n] \otimes [1] = [a']$, so this is well defined.

We moreover have $fg([a]) = f([a] \otimes [1]) = [a]$ and $gf([a], [b]) = g([ab]) = [ab] \otimes [1] = [a] \otimes [b]$. Hence f is an isomorphism. (Note that this argument works to show in general that $\mathbf{Z}/m \otimes \mathbf{Z}/n \cong \mathbf{Z}/\gcd(m, n)$). □

Exercise (1.2.H)**.** *Show that* $(-) \otimes_A N$ *is a covariant functor* $\mathsf{Mod}_A \to \mathsf{Mod}_A$, *and that it is right exact.*

Proof. Let $f : M \to M'$ be an A-module map. Then $(m, n) \to f(m) \otimes n$ is bilinear, so it yields a map $f_* : M \otimes N \to M' \otimes N$, which on generators is given by $m \otimes n \mapsto f(m) \otimes n$. This respects composition, thus making $(-) \otimes N$ into a functor.

Let $M \xrightarrow{f} M' \xrightarrow{g} M'' \xrightarrow{h} 0$ be exact. Tensoring, we get a complex

$$M \otimes N \xrightarrow{f_*} M' \otimes N \xrightarrow{g_*} M'' \otimes N \xrightarrow{h_*} 0,$$

which we need to check is exact. We know by the first isomorphism theorem that g induces an isomorphism $M'/\operatorname{im}(f) \to M''$.

Claim 1.1. Tensor products commute with quotients.

Proof. Let P, Q, R be A-modules. Suppose Q is a submodule of P. Consider the map $\tilde{f} : (P/Q) \times R \to (P \otimes R)/(Q \otimes R)$ given by $([p], r) \mapsto [p \otimes r]$. If $[p] = [p']$, then $[p \otimes r] - [p' \otimes r] = [p - p' \otimes r] = [q \otimes r] = 0$ so this map is well defined. One can also check bilinearity. Hence it extends to a map $f : (P/Q) \otimes R \to (P \otimes R)/(Q \otimes R)$.

We now define a map $\tilde{g} : P \times R \to (P/Q) \otimes R$ via $(p, r) \mapsto [p] \otimes r$. This is bilinear, hence yields a map $g' : P \otimes R \to (P/Q) \otimes R$, which further factors through the quotient by $Q \otimes R$, yielding a map $g : (P \otimes R)/(Q \otimes R) \to (P/Q) \otimes R$. And f, g are mutually inverse, yielding the claim. □

By the claim, g_* induces an isomorphism (since functors send isos to isos)

$$M'' \otimes N \leftarrow (M'/\operatorname{im}(f)) \otimes N \cong (M' \otimes N)/\operatorname{im}(f_*)$$

so $\operatorname{im}(f_*) = \ker(g_*)$. And surjectivity of g_* follows directly from surjectivity of g. □

Exercise (1.2.I)**.** *The tensor product (T, t) of two modules M, N is unique up to unique isomorphism. (Here $t : M \times N \to T$ is the bilinear map given by the universal property.)*

Proof. Let (T', t') be another pair satisfying the universal property. Then t' must factor through T, i.e. there exists a unique map $f : T \to T'$ such that $f \circ t = t'$. Likewise, there exists a unique map $g : T' \to T$ such that $g \circ t' = t$. Plugging these two formulas into each other we get $f \circ g \circ t' = t'$ and $g \circ f \circ t = t$.

Now we use the universal property again. Since t' is a bilinear map from $M \times N$ to T', it must factor uniquely through T'. One such factoring is as above, $(f \circ g) \circ t' = t'$, but another factoring is $\mathrm{id}_{T'} \circ t'$, so by uniqueness, $f \circ g = \mathrm{id}_{T'}$. Likewise, $g \circ f = \mathrm{id}_T$, hence f, g are inverses, and the unique isomorphisms between T, T'. □

Exercise (1.2.J)**.** *Show that the tensor product actually exists, i.e. the construction from §1.2.5 satisfies the universal property.*

Proof. Recall that $M \otimes N$ is the free A-module generated by $M \times N$, quotiented by the following relations:

(a) $(m_1 + m_2, n) - (m_1, n) - (m_2, n)$
(b) $(m, n_1 + n_2) - (m, n_1) - (m, n_2)$
(c) $a(m, n) - (am, n)$
(d) $a(m, n) - (m, an)$

for $a \in A, m, m_1, m_2 \in M, n, n_1, n_2 \in N$. Let $f : M \times N \to P$ be a bilinear A-module morphism. It follows by bilinearity that

(a) $f((m_1 + m_2, n) - (m_1, n) - (m_2, n)) = f(m_1, n) + f(m_2, n) - f(m_1, n) - f(m_2, n) = 0$
(b) $f((m, n_1 + n_2) - (m, n_1) - (m, n_2)) = f(m, n_1) + f(m, n_2) - f(m, n_1) - f(m, n_2) = 0$
(c) $f(a(m, n) - (am, n)) = af(m, n) - af(m, n) = 0$
(d) $f(a(m, n) - (m, an)) = af(m, n) - af(m, n) = 0$

so f factors uniquely through the quotient. □

Exercise (1.2.K)**.** *"Important exercise"*

(a) If M is an A-module and $\phi : A \to B$ is a ring morphism, give $B \otimes_A M$ the structure of a B-module. Show that this gives a functor $\mathsf{Mod}_A \to \mathsf{Mod}_B$.

Proof. Regard B as an A-module via the action $A \times B \to B : (a, b) \mapsto \phi(a)b$. Now $B \otimes_A M$ makes sense as an A-module. Now for fixed $b \in B$, consider the map $\tilde{f}_b : B \times M \to B \otimes_A M$ given by $(b', m) \to bb' \otimes m$. We compute

$$\tilde{f}_b(\phi(a)b', m) = b\phi(a)b' \otimes m = \phi(a)(bb' \otimes m) = a \cdot \tilde{f}_b(b', m)$$

and

$$\tilde{f}_b(b', am) = bb' \otimes am = a \cdot (bb' \otimes m) = a \cdot \tilde{f}_b(b', m)$$

so $\tilde{f}_b$ is A-bilinear, and it induces a map $f_b : B \otimes_A M \to B \otimes_A M$, which yields the B-module structure on $B \otimes_A M$.

To see that this is a functor, suppose we have an A-module morphism $g : M \to N$. Then we get a map $B \times M \to B \otimes_A N$ given by $(b, m) \mapsto b \otimes g(m)$. This is again A-bilinear, so it induces a map $g_* : B \otimes_A M \to B \otimes_A N$, which acts on elementary tensors via $b \otimes m \mapsto b \otimes g(m)$. □

(b) *Following notation above, if $\psi : A \to C$ is a ring morphism, then $B \otimes_A C$ has a ring structure.*

Proof. Again, C becomes an A-module via the action $(a, c) \mapsto \psi(a)c$. Then $B \otimes_A C$ automatically gets an abelian group structure. To define the ring product, note that specifying a map

$$(B \otimes C) \times (B \otimes C) \to B \otimes C$$

is the same thing as specifying a map

$$B \otimes C \to \hom_{\mathsf{Mod}_A}(B \otimes C, B \otimes C)$$

so we can start here. First define a map

$$B \times C \to \hom_{\mathsf{Mod}_A}(B \times C, B \otimes C) : (b_1, c_1) \mapsto ((b_2, c_2) \mapsto b_1 b_2 \otimes c_1 c_2).$$

By properties of the tensor product, the image of this map is an A-bilinear map, so we get a map

$$B \times C \to \hom_{\mathsf{Mod}_A}(B \otimes C, B \otimes C) : b_1, c_1 \mapsto (b_2 \otimes c_2 \mapsto b_1 b_2 \otimes c_1 c_2).$$

Again, this is also A-bilinear. For example we can check

$$\phi(a) b_1 \otimes c_1 \mapsto (b_2 \otimes c_2 \mapsto \phi(a) b_1 b_2 \otimes c_1 c_2) = (b_2 \otimes c_2 \mapsto \phi(a)(b_1 b_2 \otimes c_1 c_2)).$$

Hence we get a map

$$B \otimes C \to \hom_{\mathsf{Mod}_A}(B \otimes C, B \otimes C) : b_1 \otimes c_1 \mapsto (b_2 \otimes c_2 \mapsto b_1 b_2 \otimes c_1 c_2)$$

which is equivalent to the map

$$\mu : (B \otimes C) \times (B \otimes C) \to B \otimes C : (b_1 \otimes c_1), (b_2 \otimes c_2) \mapsto b_1 b_2 \otimes c_1 c_2$$

which is A-bilinear by definition since it comes from the universal property. Now for $x, y, z \in B \otimes C$, we have by bilinearity

$$\mu(x + y, z) = \mu(x, z) + \mu(y, z), \quad \mu(x, y + z) = \mu(x, y) + \mu(x, z)$$

so this satisfies distributivity. The identity element is $1_B \otimes 1_C$, and μ is obviously associative, so we are done. □

Exercise (1.2.L). *If S is a multiplicative subset of A and M is an A-module, describe a natural isomorphism $(S^{-1}A) \otimes_A M \to S^{-1}M$ (as both A-modules and $S^{-1}A$-modules).*

Proof. Consider first the map $S^{-1}A \times M \to S^{-1}M$ given by $(a/s, m) \mapsto (a \cdot m)/s$. This is A-bilinear since

$$(a' \cdot a/s, m) = (a'a/s, m) \mapsto (aa' \cdot m)/s = a \cdot (a' \cdot m)/s$$

and

$$(a/s, a' \cdot m) \mapsto (a \cdot (a' \cdot m))/s = (aa' \cdot m)/s = a \cdot (a' \cdot m)/s.$$

Hence this induces a map $f : (S^{-1}A) \otimes_A M \to S^{-1}M$ which acts on elementary tensors as

$$f(a/s \otimes m) = (a \cdot m)/s.$$

We can define an inverse map as follows. First consider the map $\tilde{g} : M \to (S^{-1}A) \otimes_A M$ via $m \mapsto 1/1 \otimes m$. Since multiplication by elements of S are invertible in $(S^{-1}A) \otimes_A M$, the map $\tilde{g}$ factors through the localization to yield a map $g : S^{-1}M \to (S^{-1}A) \otimes_A M$ which acts via $m/s \mapsto s^{-1}(1/1 \otimes m) = (1/s \otimes m)$. Then we can check

$$gf(a/s \otimes m) = g((a \cdot m)/s) = (1/s \otimes a \cdot m) = (a/s \otimes m)$$

and

$$fg(m/s) = f(1/s \otimes m) = m/s.$$

It remains to check that f is also a $S^{-1}A$-module map. Recall from Exercise 1.2.K.1 that $S^{-1}A \otimes_A M$ has the structure of a $S^{-1}A$-module via the map $A \to S^{-1}A$, $a \mapsto a/1$. So for $a/s, a'/s' \in S^{-1}A$, $m \in M$, we have

$$f(a/s \cdot (a'/s') \otimes m) = f(aa'/ss' \otimes m) = (aa' \cdot m)/ss' = (a'/s') \cdot (am/s)$$

as desired. For naturality, we need to check that the following diagram commutes for any map of A-modules $\phi : M \to N$:

$$\begin{tikzcd}
(S^{-1}A) \otimes_A M \arrow[r, "f"] \arrow[d, "\phi_*"'] & S^{-1}M \arrow[d, "\phi_*"] \\
(S^{-1}A) \otimes_A N \arrow[r, "f"'] & S^{-1}N
\end{tikzcd}$$

but this is not too bad, for following the top and right arrows we have

$$\phi_*(f(a/s \otimes m)) = \phi_*((a \cdot m)/s) = \phi(a \cdot m)/s$$

and following the left and bottom ones we have

$$f(\phi_*(a/s \otimes m)) = f(a/s \otimes \phi(m)) = (a \cdot \phi(m))/s = \phi(a \cdot m)/s$$

as desired. □

Exercise (1.2.M)**.** *Show that tensor products commute with arbitrary direct sums, i.e. for A-modules M and $\{N_\alpha\}_{\alpha \in \Lambda}$, define an isomorphism $M \otimes (\oplus_{\alpha \in \Lambda} N_\alpha) \to \oplus_{\alpha \in \Lambda}(M \otimes N_\alpha)$.*

Proof. As with every map out of a tensor prouduct, we begin by defining a map

$$\tilde{f} : M \times (\oplus_{\alpha \in \Lambda} N_\alpha) \to \oplus_{\alpha \in \Lambda}(M \otimes N_\alpha), (m, (n_\alpha)_{\alpha \in \Lambda}) \mapsto (m \otimes n_\alpha)_{\alpha \in \Lambda}.$$

This, being bilinear, induces a map $f : M \otimes (\oplus_{\alpha \in \Lambda} N_\alpha) \to \oplus_{\alpha \in \Lambda}(M \otimes N_\alpha)$ which acts on elementary tensors via $m \otimes (n_\alpha)_{\alpha \in \Lambda} \mapsto (m \otimes n_\alpha)_{\alpha \in \Lambda}$.

For each $\alpha \in \Lambda$, consider the map

$$\tilde{g}_\alpha : M \times N_\alpha \to M \otimes (\oplus_{\alpha \in \Lambda} N_\alpha), (m, n) \mapsto m \otimes \iota_\alpha(n)$$

where $\iota_\alpha : N_\alpha \to \oplus_\alpha N_\alpha$ is the inclusion. This is A-bilinear, so it induces a map $g_\alpha : M \otimes N_\alpha \to M \otimes (\oplus_{\alpha \in \Lambda} N_\alpha)$. The sum of all the g_α yields a map $g := \oplus_{\alpha \in \Lambda}(M \otimes N_\alpha) \to M \otimes (\oplus_{\alpha \in \Lambda} N_\alpha)$.

And we check

$$\begin{aligned}
fg((m_\alpha \otimes n_\alpha)_{\alpha \in \Lambda}) &= f \sum_{\alpha \in \Lambda} (m_\alpha \otimes \iota_\alpha(n_\alpha)) \\
&= \sum_{\alpha \in \Lambda} f(m_\alpha \otimes \iota(n_\alpha)) \\
&= \sum_{\alpha \in \Lambda} (\ldots, 0, m_\alpha \otimes n_\alpha, 0, \ldots) \\
&= (m_\alpha \otimes n_\alpha)_{\alpha \in \Lambda}
\end{aligned}$$

and

$$gf(m \otimes (n_\alpha)_{\alpha\in\Lambda}) = g((m \otimes n_\alpha)_{\alpha\in\Lambda}) = \sum_{\alpha\in\Lambda} m \otimes \iota_\alpha(n_\alpha) = m \otimes \sum_{\alpha\in\Lambda} \iota_\alpha(n_\alpha) = m \otimes (n_\alpha)_{\alpha\in\Lambda}$$

so f is an isomorphism. □

Exercise (1.2.H)**.** *Show that in* Set*, the fiber product is given by* $X \times_Z Y = \{(x, y) \in X \times Y : \alpha(x) = \beta(y)\}$.

Proof. Call the right hand side of above R. Let W be a set with maps $f_X : W \to X$ and $f_Y : W \to Y$ such that $\alpha \circ f_X = \beta \circ f_Y$. Define the map $\phi : W \to R$ via $\phi(w) = (f_X(w), f_Y(w))$. This actually lands in R since $\alpha \circ \pi_X(\phi(w)) = \alpha \circ f_X(w) = \beta \circ f_Y(w) = \beta \circ \pi_Y(\phi(w))$. Then clearly $\pi_X \circ \phi = f_X$, and $\pi_Y \circ \phi = f_Y$. Thus R satisfies the universal property so it is *the* fiber product in the category of sets. □

Exercise (1.2.O)**.** *If X is a topological space, show that fiber products always exist in the category of open sets of X by describing what a fibered product is.*

Proof. Remember that the morphisms here are just inclusions of open sets. So suppose we have open sets U, U' mapping into W, i.e. $U, U' \subset W$. We claim that the fiber product here is just $U \cap U'$ (intersections of two open sets are open), with the morphisms given by inclusions into U, U' . To see this, given another open set V with morphisms $V \to U$ and $V \to U'$ that agree with $U \to W$ and $U' \to W$, then we know that V must be a subset of both U and U'. It follows that V is a subset of $U \cap U'$. And by definition, all the morphisms are inclusions, so everything commutes. □

Exercise (1.2.P)**.** *If Z is the final object in the category $\mathcal{C}$, show that the fiber product $X \times_Z Y$ is given by the product $X \times Y$. (Assume all (fiber) products exist.)*

Proof. Let $W \in \mathcal{C}$ with maps $f_X : W \to X$ and $f_Y : W \to Y$ as in the setup of the universal property. Then by the universal property of products, we get a map $\phi : W \to X \times Y$ which factors f_X and f_Y. Moreover, since Z is the final object, the compositions $X \times Y \to X \to Z$ and $X \times Y \to Y \to Z$ must agree. □

Exercise (1.2.Q)**.** *Stacking Cartesian squares gives a Cartesian rectangle:*

$$\begin{tikzcd}
U \arrow[r] \arrow[d] & V \arrow[d] \\
W \arrow[r] \arrow[d] & X \arrow[d] \\
Y \arrow[r] & Z
\end{tikzcd}$$

Proof. Suppose we have maps $f_Y : A \to Y$ and $f_V : A \to V$ such that their postcompositions with $Y \to Z$ and $V \to Z$ agree. Then since the bottom square is Cartesian, there is a unique map $f_{AW} : A \to W$ which factors f_Y and $f_{VX} \circ f_V$, i.e. $f_{WX} \circ f_{AW} = f_{VX} \circ f_V$. Now since the top square is Cartesian, there is a unique map $f_{AW} : A \to U$ factoring f_{AW} (hence f_Y) and f_V, completing the proof. See below for a figure of all the maps involved:

$$\begin{tikzcd}
A \arrow[rrd, bend left, "f_V"] \arrow[rd, dashed, "f_{AU}"] \arrow[rdd, dashed, "f_{AW}"'] \arrow[rddd, bend right, "f_Y"'] & & \\
& U \arrow[r] \arrow[d] & V \arrow[d, "f_{VX}"] \\
& W \arrow[r, "f_{WX}"] \arrow[d] & X \arrow[d] \\
& Y \arrow[r] & Z
\end{tikzcd}$$

□

Exercise (1.2.R)**.** *Given morphisms $X_1 \to Y$, $X_2 \to Y$, and $Y \to Z$, there is a natural morphism $X_1 \times_Y X_2 \to X_1 \times_Z X_2$ (assuming both fiber products exist).*

Proof. Consider the following diagram:

$$\begin{tikzcd}
X_1 \times_Y X_2 \arrow[rrrd, bend left] \arrow[rd, dashed] \arrow[rddd, bend right] & & & \\
& X_1 \times_Z X_2 \arrow[rr] \arrow[dd] & & X_2 \arrow[d] \\
& & & Y \arrow[d] \\
& X_1 \arrow[r] & Y \arrow[r] & Z
\end{tikzcd}$$

where the outside arrows commute by definition. Hence by the universal property we get the dotted morphism. □

Exercise (1.2.S)**.** *Show that the diagonal-base-change diagram below is Cartesian:*

$$\begin{tikzcd}
X_1 \times_Y X_2 \arrow[r] \arrow[d] & X_1 \times_Z X_2 \arrow[d] \\
Y \arrow[r] & Y \times_Z Y
\end{tikzcd}$$

Proof. The top arrow is defined by the diagram from the previous exercise. The left arrow is either composition $X_1 \times_Y X_2 \to X_i \to Y$. The right arrow is the dotted arrow below:

$$\begin{tikzcd}
X_1 \times_Y X_2 \arrow[r] \arrow[d] \arrow[rd, dashed] & X_2 \arrow[rd] & \\
X_1 \arrow[rd] & Y \times_Z Y \arrow[r] \arrow[d] & Y \arrow[d] \\
& Y \arrow[r] & Y
\end{tikzcd}$$

And the bottom arrow is the diagonal map. Stacking the top and right arrow diagrams in front of each other, padding the middle, we get a diagram as follows:

$$
\begin{tikzcd}
X_1 \times_Y X_2 \arrow[dr, dashed, red] \arrow[rrrr, bend left, blue] \arrow[dddr, bend right, blue] & & & & & & \\
& X_1 \times_Z X_2 \arrow[rrr] \arrow[ddd] \arrow[dr, red, "\mathrm{id}"] & & & X_2 \arrow[ddd] \arrow[dr, blue, "\mathrm{id}"] & & \\
& & X_1 \times_Z X_2 \arrow[rrr] \arrow[ddd] \arrow[dr, dashed, red] & & & X_2 \arrow[dr, blue] & \\
& & & Y \times_Z Y \arrow[rrr] \arrow[ddd] & & & Y \arrow[ddd] \\
& X_1 \arrow[rrr] \arrow[dr, blue, "\mathrm{id}"] & & & Z \arrow[ddrr] & & \\
& & X_1 \arrow[dr, blue] & & & & \\
& & & Y \arrow[rrr] & & & Z
\end{tikzcd}
$$

So the composition of the red arrows is the unique map coming from the universal property applied to the front square and composition of the blue arrows. On the other hand, if we stack the left and bottom arrow diagrams on top of each other, padding the middle, we get

$$
\begin{tikzcd}
X_1 \times_Y X_2 \arrow[rr, bend left, blue] \arrow[dd, bend right, blue] \arrow[dr, dashed, red] & & X_2 \arrow[drr, bend left, blue] & & & & \\
& Y \times_Z Y \arrow[rrr] \arrow[ddd] \arrow[dr, red] & & & Y \arrow[ddd] \arrow[dr, blue] & & \\
X_1 \arrow[ddr, bend right, blue] & & Y \arrow[rrr] \arrow[ddd] \arrow[dr, dashed, red] & & & Y \arrow[dr, blue] & \\
& & & Y \times_Z Y \arrow[rrr] \arrow[ddd] & & & Y \arrow[ddd] \\
& Y \arrow[rrr] \arrow[dr, blue] & & & Z \arrow[ddrr] & & \\
& & Y \arrow[dr, blue] & & & & \\
& & & Y \arrow[rrr] & & & Z
\end{tikzcd}
$$

But now the blue arrows on the outside are the same, so by the universal property, the composition of the red arrows in both diagrams must agree, so the square in the exercise commutes.

Now let A be an object with morphisms to Y and $X_1 \times_Z X_2$ such that their postcompositions with the according maps are equal, i.e. the following diagram commutes:

$$\begin{tikzcd} A \arrow[drr, bend left] \arrow[ddr, bend right] & & \\ & X_1 \times_Y X_2 \arrow[r] \arrow[d] & X_1 \times_Z X_2 \arrow[d] \\ & Y \arrow[r] & Y \times_Z Y \end{tikzcd}$$

There being a morphism $A \to X_1 \times_Z X_2$ implies that we have morphisms $f_i : A \to X_i$ for $i = 1, 2$, whose postcompositions with $X_i \to Y \to Z$ are equal. The right arrow above is induced by maps $X_1 \times_Z X_2 \to X_i \to Y$, which both have to agree with $A \to Y$, that is they are equal. Hence the compositions $A \to X_i \to Y$ induce a map $A \to X_1 \times_Y X_2$, completing the proof. □

Exercise (1.2.T)**.** *Show that the coproduct in the category of sets is disjoint union.*

Proof. Let X, Y be sets and let us take the disjoint union to be

$$X \coprod Y := \{(x, 0) : x \in X\} \cup \{(y, 1) : y \in Y\}.$$

Define the inclusions $\iota_X : X \to X \coprod Y$ via $x \mapsto (x, 0)$ and $\iota_Y : Y \to X \coprod Y$ via $y \mapsto (y, 1)$. Let A be a set with maps $f_X : X \to A$, $f_Y : Y \to A$. Then we can define a map $f : X \coprod Y \to A$ via $(x, 0) \mapsto f_X(x)$ and $(y, 1) \mapsto f_Y(y)$, which makes everything commute. For uniqueness, if $g : X \coprod Y \to A$ is another map making things commute, we have

$$g(x, 0) = g(\iota_X(x)) = f_X(x) = f(x, 0)$$

and likewise for Y. □

Exercise (1.2.U)**.** *For ring morphisms $A \to B$, $A \to C$, show that there is a natural morphism $B \to B \otimes_A C$ given by $b \mapsto b \otimes 1$ and likewise $C \to B \otimes_A C$ given by $c \mapsto 1 \otimes c$. Show that these maps make the following a cocartesian diagram in the category of rings.*

$$\begin{tikzcd} A \arrow[r, "f_B"] \arrow[d, "f_C"'] & B \arrow[d] \\ C \arrow[r] & B \otimes_A C \end{tikzcd}$$

Proof. To see that $b \to b \otimes 1$ is a ring map, we have $b + b' \mapsto (b + b') \otimes 1 = b \otimes 1 + b' \otimes 1$, and $bb' \mapsto bb' \otimes 1 = (b \otimes 1)(b' \otimes 1)$, and $1 \mapsto 1 \otimes 1$. Likewise for $C \to B \otimes_A C$. The square commutes since $f_B(a) \otimes 1 = f_B(a) \cdot 1 \otimes 1 = 1 \otimes f_C(a) \cdot 1 = 1 \otimes f_C(a)$.

Let D be another ring with maps $g_B : B \to D$ and $g_C : C \to D$ such that $g_B f_b = g_C f_C$. Define a map $B \otimes_A C \to D$ beginning with $B \times C \to D, (b, c) \mapsto g_B(b) g_C(c)$. By virtue of g_B being a ring morphism, we have

$$(f_B(a)b, c) \mapsto g_B(f_B(a)b) g_C(c) = g_B(f_B(a)) g_B(b) g_C(c)$$

and likewise in the second argument, so this map is A-bilinear. Consequently we have a map $\phi : B \otimes_A C \to D$ acting on basis elements via $b \otimes c \mapsto g_B(b) g_C(c)$. Then $g_B(b) = g_B(b)1 = g_B(b) g_C(1) = \phi(b \otimes 1)$, and likewise $g_C(c) = \phi(1 \otimes c)$, so we are done. □

Exercise (1.2.V)**.** *Show that the composition of two monomorphisms is a monomorphism.*

Proof. Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be monomorphisms. Suppose $\pi_1, \pi_2 : W \to X$ satisfy $(g \circ f) \circ \pi_1 = (g \circ f) \circ \pi_2$. Then by associativity of composition, we have $g \circ (f \circ \pi_1) = g \circ (f \circ \pi_2)$, and because g is a mono, $f \circ \pi_1 = f \circ \pi_2$, and because f is a mono, $\pi_1 = \pi_2$. □

Exercise (1.2.W)**.** *Show that a morphism $\pi : X \to Y$ is a monomorphism if and only if the fiber product $X \times_Y X$ exists and the induced diagonal map $\delta_\pi : X \to X \times_Y X$ is an isomorphism.*

Proof. Forward direction. We claim that the fiber product is given by any object X' with a chosen isomorphism $\psi : X' \to X$ serving as both projection maps. This exists, for example, by taking $X' = X$ and $\psi = \mathrm{id}$. To see this is the fiber product, it is clear that the square commutes. Given any maps $f_1, f_2 : W \to X$ such that $\pi \circ f_1 = \pi \circ f_2$, we know that $f_1 = f_2$ since π is a monomorphism. Then defining $W \to X$ by $\psi^{-1} \circ f_1$ makes things commute. If (X'', ψ'') is another fiber product, then $\psi''^{-1} \circ \psi$ gives the unique isomorphism, so (X', ψ) satisfies the universal property. Finally, the diagonal map must satisfy $\psi \circ \delta_\pi = \mathrm{id}$, so it is an isomorphism since ψ is an isomorphism.

Backward direction. By assumption, δ_π is an isomorphism, and it satisfies the identity $\mathrm{pr}_X \circ \delta_\pi = \mathrm{id}$, so both projection maps pr_X are inverse to δ_π, so both projection maps are the same. Then given maps $f_1, f_2 : W \to X$ such that $\pi \circ f_1 = \pi \circ f_2$, by the the universal property of fiber products we get a unique map $\phi : W \to X \times_Y X$ such that $f_1 = \delta_\pi^{-1} \circ \phi = f_2$. □

Exercise (1.2.X)**.** *Given morphisms $X_1 \to Y$, $X_2 \to Y$, and a monomorphism $Y \to Z$, show that $X_1 \times_Y X_2 \to X_1 \times_Z X_2$ is an isomorphism.*

Proof. By the previous exercise, g in the diagonal-base-change diagram below is an isomorphism

$$\begin{tikzcd} X_1 \times_Y X_2 \arrow[r, "f"] \arrow[d, "h"'] & X_1 \times_Z X_2 \arrow[d, "j"] \\ Y \arrow[r, "g"'] & Y \times_Z Y \end{tikzcd}$$

We can proceed by showing the more general fact that in an arbitrary Cartesian square

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "h"'] & B \arrow[d, "j"] \\ C \arrow[r, "g"] & D \end{tikzcd}$$

that if g is an isomorphism then so is f. The trick is to first consider the square

$$\begin{tikzcd} B \arrow[r, "\mathrm{id}"] \arrow[d, "g^{-1} \circ j"'] & B \arrow[d, "j"] \\ C \arrow[r, "g"] & D \end{tikzcd}$$

which we claim is Cartesian. It obviously commutes. Given maps $\phi_B : X \to B$ and $\phi_C : X \to C$ such that $j \circ \phi_B = g \circ \phi_C$, the map we want $X \to B$ is just given by ϕ_B, since $\phi_B = \mathrm{id} \circ \phi_B$, and $g^{-1} \circ j \circ \phi_B = g^{-1} \circ g \circ \phi_C = \phi_C$. So B is another fiber product, and by the universal property there is an isomorphism $u : A \to B$ making the following diagram commute

$$\begin{tikzcd} A \arrow[dr, dashed, "u"] \arrow[drr, bend left, "f"] \arrow[ddr, bend right, "h"'] & & \\ & B \arrow[r, "\mathrm{id}"'] \arrow[d, "g^{-1} \circ j"] & B \arrow[d, "h"] \\ & C \arrow[r, "g"'] & D \end{tikzcd}$$

Because f also makes the diagram commute, $u = f$ and we are done. □

Exercise (1.2.Y)**.** *Yoneda lemma.*

(a) *Let A, A' be objects in a category $\mathcal{C}$. Suppose there exist a collection of maps $i_C : \mathrm{Hom}(C, A) \to \mathrm{Hom}(C, A')$ for $C \in \mathcal{C}$ that commute with the maps $\mathrm{Hom}(Z, X) \to \mathrm{Hom}(Y, X)$ induced by morphisms $Y \to Z$. (i.e. a natural transformation between the functors $\mathrm{Hom}(-, A)$ and $\mathrm{Hom}(-, A')$). Show that the collection of maps $\{i_C\}$ are induced by a unique morphism $g : A \to A'$, i.e. each i_C is of the form $u \mapsto g \circ u$.*

Proof. Consider plugging in $C = A$. Then we have a map $i_A : \mathrm{Hom}(A, A) \to \mathrm{Hom}(A, A')$. We know that $\mathrm{id}_A \in \mathrm{Hom}(A, A)$, and we claim that $g = i_A(\mathrm{id}_A)$. Now suppose we have a morphism $\phi : C \to A$. By the naturality/commutativity assumption, this gives rise to a commutative diagram

$$\begin{tikzcd} \mathrm{Hom}(A, A) \arrow[r, "i_A"] \arrow[d, "-\circ\phi"'] & \mathrm{Hom}(A, A') \arrow[d, "-\circ\phi"] \\ \mathrm{Hom}(C, A) \arrow[r, "i_C"'] & \mathrm{Hom}(C, A') \end{tikzcd}$$

Tracing through the element id_A, the top-right path ends up at $i_A(\mathrm{id}_A) \circ \phi = g \circ \phi$, while the bottom-left path ends up at $i_C(\mathrm{id}_A \circ \phi) = i_C(\phi)$. It follows that $i_C(\phi) = g \circ \phi$, so we are done. □

(b) *If furthermore the i_C are all bijections, show that g is an isomorphism. (So if we have a natural isomorphism of the functors $\mathrm{Hom}(-, A)$ and $\mathrm{Hom}(-, A')$ then we also get an isomorphism $A \to A'$.)*

Proof. Since $i_{A'}$ is an isomorphism, there is some $f \in \mathrm{Hom}(A', A)$ such that $i_{A'}(f) = \mathrm{id}_{A'}$. Now consider the same commutative diagram as above with $C = A'$. Let the vertical arrows be precomposition by f. Then commutativity yields $g \circ f = i_A(\mathrm{id}_A) \circ f = \mathrm{id}_{A'}$.

On the other hand, flip this commutative diagram diagonally and swap the direction of the horizontal arrows to get

$$\begin{tikzcd} \mathrm{Hom}(A', A') \arrow[r, "i_{A'}^{-1}"] \arrow[d, "-\circ g"'] & \mathrm{Hom}(A', A) \arrow[d, "-\circ g"'] \\ \mathrm{Hom}(A, A') \arrow[r, "i_A^{-1}"] & \mathrm{Hom}(A, A) \end{tikzcd}$$

which yields $f \circ g$ following the top and right paths and id_A following the bottom left. So g is an isomorphism. □

Exercise (1.2.Z)**.** *Yoneda lemma again.*

(a) *Let A, B be objects in a category $\mathcal{C}$. Give a bijection between $\mathrm{Hom}(\mathrm{Hom}(A, -), \mathrm{Hom}(B, -))$ and $\mathrm{Hom}(B, A)$.*

Solution. (This is Exercise 1.2.Z.c, take $F = \mathrm{Hom}(B, -)$.) For the exact same proof, read on. Given a natural transformation $\alpha \in \mathrm{Hom}(\mathrm{Hom}(A, -), \mathrm{Hom}(B, -))$, define $\Phi(\alpha) = \alpha_A(\mathrm{id}_A) \in \mathrm{Hom}(B, A)$. Conversely, given $g \in \mathrm{Hom}(B, A)$, define a natural transformation by $\Psi(g)_C(f) = f \circ g$. This is indeed natural since given any morphism $\varphi : C \to C'$, the square

$$\begin{tikzcd} \mathrm{Hom}(A, C) \arrow[r, "-\circ g"] \arrow[d, "\varphi\circ-"'] & \mathrm{Hom}(B, C) \arrow[d, "\varphi\circ-"'] \\ \mathrm{Hom}(A, C') \arrow[r, "-\circ g"'] & \mathrm{Hom}(B, C') \end{tikzcd}$$

commutes by associativity of morphism composition.

Now we need to check that Φ and Ψ are inverses. In one direction, given a natural transformation α, $C \in \mathcal{C}$, and $f \in \mathrm{Hom}(A, C)$, we have

$$(\Psi \circ \Phi(\alpha))_C(f) = (\Psi(\alpha_A(\mathrm{id}_A)))_C(f) = f \circ \alpha_A(\mathrm{id}_A).$$

By naturality of α, the square

$$\begin{tikzcd} \mathrm{Hom}(A,A) \arrow[r, "\alpha_A"] \arrow[d, "f\circ -"'] & \mathrm{Hom}(B,A) \arrow[d, "f\circ -"] \\ \mathrm{Hom}(A,C) \arrow[r, "\alpha_C"'] & \mathrm{Hom}(B,C) \end{tikzcd}$$

yields the equality $f \circ \alpha_A(\mathrm{id}_A) = \alpha_C(f)$, which shows $\Psi \circ \Phi$ is the identity.

In the other direction, let $g \in \mathrm{Hom}(B, A)$. We compute

$$\Phi(\Psi(g)) = \Psi(g)_A(\mathrm{id}_A) = \mathrm{id}_A \circ g = g$$

which completes the proof. □

(b) State and prove the dual statement for $\mathrm{Hom}(\mathrm{Hom}(-, A), \mathrm{Hom}(-, B))$ *and* $\mathrm{Hom}(A, B)$.

Proof. Let $\Phi : \mathrm{Hom}(\mathrm{Hom}(-, A), \mathrm{Hom}(-, B)) \to \mathrm{Hom}(A, B)$ be the map $\alpha \mapsto \alpha_A(\mathrm{id}_A)$ from Exercise 1.2.Y.a. Conversely given $g \in \mathrm{Hom}(A, B)$ define a natural transformation $\Psi(g)$ with $\Psi(g)_C(f) = g \circ f$ where $f \in \mathrm{Hom}(C, A)$. Then, given a natural transformation α, $C \in \mathcal{C}$, and $f \in \mathrm{Hom}(C, A)$, we have

$$\Psi(\Phi(\alpha))_C(f) = \Psi(\alpha_A(\mathrm{id}_A))_C(f) = \alpha_A(\mathrm{id}_A) \circ f.$$

Again, by naturality we get a commutative diagram

$$\begin{tikzcd} \mathrm{Hom}(A,A) \arrow[r, "\alpha_A"] \arrow[d, "-\circ f"'] & \mathrm{Hom}(A,B) \arrow[d, "-\circ f"] \\ \mathrm{Hom}(C,A) \arrow[r, "\alpha_C"'] & \mathrm{Hom}(C,B) \end{tikzcd}$$

which yields the equality $\alpha_A(\mathrm{id}_A) \circ f = \alpha_C(f)$, so $\Psi(\Phi(\alpha)) = \alpha$. In the other direction, for $g \in \mathrm{Hom}(A, B)$, we have

$$\Phi(\Psi(g)) = (\Psi(g))_A(\mathrm{id}_A) = g \circ \mathrm{id}_A = g$$

which completes the proof!!!!!! □

(c) Let $F : \mathcal{C} \to \mathsf{Set}$ *be a covariant functor. Give a bijection between* $\mathrm{Hom}(\mathrm{Hom}(A, -), F)$ *and* FA.

Solution. We do the exact same thing as before. In one direction, given a natural transformation $\alpha : \mathrm{Hom}(A, -) \Rightarrow F$, define $\Phi(\alpha)$ to be $\alpha_A(\mathrm{id}_A)$. In the other direction, given $X \in FA$, $C \in \mathcal{C}$ and $f \in \mathrm{Hom}(A, C)$, define $\Psi(X)_C(f) = (Ff)(X)$.

We check, again, that given a natural transformation $\alpha : \mathrm{Hom}(A, -) \Rightarrow F$, $C \in \mathcal{C}$ and $f \in \mathrm{Hom}(A, C)$, that

$$\Psi(\Phi(\alpha))_C(f) = \Psi(\alpha_A(\mathrm{id}_A))_C(f) = (Ff)(\alpha_A(\mathrm{id}(A)))$$

which by naturality is equal to $\alpha_C(f)$. And on the other hand, given an element $X \in FA$, we have

$$\Phi(\Psi(X)) = \Psi(X)_A(\mathrm{id}_A) = (F\,\mathrm{id}_A)(X) = \mathrm{id}_{FA}(X) = X$$

which completes the proof. □

1.3. **Limits and colimits.**

Exercise (1.3.A). *Show that if $\mathcal{I}$ has an initial object e, then the limit of any diagram $F : \mathcal{I} \to \mathcal{C}$ exists.*

Proof. We claim that the limit is just given by Fe, with the structure maps $f_j : Fe \to Fj$ given by the image of the unique morphism $e \to j$ under F, say g_j. Then for a morphism $m : j \to k$ in $\mathcal{I}$, since e is initial, we know that $m \circ g_j = g_k$, whence it follows that $Fm \circ f_j = Fm \circ Fg_j = F(m \circ Fg_j) = Fg_k = f_k$.

To see that Fe is final with respect to this property, for any $W \in \mathcal{C}$ with maps $w_j : W \to A_j$ commuting with the relevant things, we immediately have a morphism $w_e : W \to Fe$, and by assumption $w_j = f_j \circ w_e$. □

Exercise (1.3.B). *Solve 1.2.S by identifying both $X_1 \times_Y X_2$ and $Y \times_{(Y \times_Z Y)} (X_1 \times_Z X_2)$ as the limit of the diagram*

$$\begin{tikzcd} X_1 \arrow[dr] & & \\ & Y \arrow[r] & Z \\ X_2 \arrow[ur] & & \end{tikzcd} \tag{1.1}$$

Proof. First to see how this implies the result of 1.2.S, note that the long fiber product $Y \times_{(Y \times_Z Y)} (X_1 \times_Z X_2)$ is the limit of the diagram

$$\begin{tikzcd} & X_1 \times_Z X_2 \arrow[d] \\ Y \arrow[r] & Y \times_Z Y \end{tikzcd} \tag{1.2}$$

so if we show that $X_1 \times_Y X_2$ and this limit coincide, then the square

$$\begin{tikzcd} X_1 \times_Y X_2 \arrow[r] \arrow[d] & X_1 \times_Z X_2 \arrow[d] \\ Y \arrow[r] & Y \times_Z Y \end{tikzcd}$$

must be Cartesian. Now onto the proof. It is clear that $X_1 \times_Y X_2$ is the limit of diagram (1.1): the structure maps to X_i are given by the projections from the fiber product, and the structure maps to Y and Z are given by postcomposing with the provided maps $X_i \to Y$, $Y \to Z$. And by the universal property of fiber products, $X_1 \times_Y X_2$ is final w.r.t. this.

Now we show that $K := Y \times_{(Y \times_Z Y)} (X_1 \times_Z X_2)$ is also the limit of (1.1). Since it is a fiber product, it comes with a map to $X_1 \times_Z X_2$, which is the data of maps $f_i : K \to X_i$ commuting with $X_i \to Z$, as well as a map $f_y : K \to Y$. The map $f_z : K \to Z$ is given by postcomposing with $Y \to Z$. By defintion of all these fiber products, f_i, f_y, f_z must commute with each other. To see K is final w.r.t. this property, suppose we have a maps g_i from W to X_i, Y, Z which commute with the diagram (1.1). Taking the maps to X_i and Z, we obtain a map $W \to X_1 \times_Z X_2$ by the universal property of the fiber product. On the other hand, we are already given a map $W \to Y$. These fit into the following diagram:

$$\begin{tikzcd}
W \arrow[dr, dashed] \arrow[rrr, bend left] \arrow[drr, bend left] \arrow[ddr, bend right] & & & \\
& X_1 \times_Z X_2 \arrow[r] \arrow[d] & X_2 \arrow[r] \arrow[d] & Y \arrow[ddr] \\
& X_1 \arrow[r] \arrow[d] & Y \arrow[ur] \arrow[dl] \arrow[dr, dashed] & \\
& Y \arrow[drrr] & & Y \times_Z Y \arrow[ll] \arrow[uu] \\
& & & & Z
\end{tikzcd}$$

Where the arrows in blue yield an arrow from W to K. □

Exercise (1.3.C)**.** *In the category of sets, the limit of a diagram $F : \mathcal{I} \to A$ is given by*

$$L := \{(a_i)_{i\in\mathcal{I}} \in \prod_{i\in I} A_i : F(m)(a_j) = a_k \text{ for all } m \in \mathrm{Hom}(j,k)\}$$

with the standard projection maps $\pi_i : L \to A_i$.

Proof. By definition of L, given $a = (a_i)_{i\in\mathcal{I}} \in L$ and $m : j \to k$, we have by definition of L that $Fm(\pi_j(a)) = Fm(a_j) = a_k = \pi_k(a)$. Hence things commute. To show that L is final, suppose we have a family of functions $g_i : W \to A_i$ commuting with things. Then we can define $g : W \to L$ given by $w \mapsto (g_i(w))_{i\in\mathcal{I}}$. Then $\pi_j(g(w)) = g_j(w)$ for $j \in \mathcal{I}$, completing the proof. □

Exercise (1.3.D)**.** *Colimits.*

(a) *Interpret the statement* $\mathbf{Q} = \mathrm{colim}(\frac{1}{n}\mathbf{Z})$.

Solution. Consider the category of natural numbers $\mathbf{N}$ with morphisms given by divisibility, i.e. there is a morphism $a \to b$ if a divides b. Consider the functor F from $\mathbf{N}$ to abelian groups given by $n \mapsto \frac{1}{n}\mathbf{Z}$. Then if there is an arrow $n \to m$, i.e. $nk = m$ for some k, we define a homomorphism $\phi_{nm} : \frac{1}{n}\mathbf{Z} \to \frac{1}{m}\mathbf{Z}$ via $x/n \mapsto xk/m$. Thus F is actually a functor, and we claim that $\mathbf{Q}$ is the colimit of this diagram.

To see this, note that we have maps $\iota_n : \frac{1}{n}\mathbf{Z} \to \mathbf{Q}$ given by inclusion. These commute with the morphisms in $\mathbf{N}$, again with n, m, k as above we have $\iota_n(x/n) = x/n = xk/nk = xk/m = \iota_m(\phi_{nm}(x/n))$. Given another group K with morphisms $g_n : \frac{1}{n}\mathbf{Z} \to K$ for all n commuting with the divisibility maps, we can define a map $\phi : \mathbf{Q} \to K$ via $q = a/b \mapsto g_b(q)$.

This is well defined, since if $a/b = a'/b'$, then the following diagram commutes

$$\begin{tikzcd}
\frac{1}{b}\mathbf{Z} \arrow[dr, "a/b\mapsto ab'/bb'"] \arrow[drr, bend left, "g_b"] & & \\
& \frac{1}{bb'}\mathbf{Z} \arrow[r, "g_{bb'}"'] & K \\
\frac{1}{b'}\mathbf{Z} \arrow[ur, "a'/b'\mapsto a'b/bb'"] \arrow[urr, bend right, "g_{b'}"'] & &
\end{tikzcd}$$

implying that $\phi(a/b) = g_b(a/b) = g_{b'}(a'/b') = \phi(a'/b')$. And ϕ trivially commutes with the ι_n since we have $\phi(\iota_b(a/b)) = \phi(a/b) = g_b(a/b)$ by definition. □

(b) *Interpret the union of subsets of a given set as a colimit.*

Solution. Let A be a set and consider the category of subsets 2^A of A, with a morphism $X \to Y$ if $X \subset Y$. Let $\{A_i\}_{i \in I}$ be the collection of subsets we want to take the union of, interpreted as a subcategory of the category of subsets of A. Then we get a diagram from this subcategory to 2^A via inclusion, and we claim that $\cup A_i$ is the colimit of this diagram. The commutativity is obvious, if $A_j \subset A_k$, then $A_j \subset A_k \subset \cup A_i$ gives the same thing as $A_j \subset \cup A_i$. If we have another subset of A, say K with maps from each A_i commuting with inclusion, this means that $\cup A_i$ is a subset of K, automatically giving us the desired extension. Again all the morphisms commute since they are just inclusions. □

Exercise (1.3.E)**.** *Let $\mathcal{I}$ be a filtered index category. Prove that any colimit $F : \mathcal{I} \to \mathsf{Set}$ exists, and is given by*

$$K := \left\{(a_i, i) \in \coprod_{i \in \mathcal{I}} A_i\right\} \Big/ \sim$$

where the equivalence relation is

$$(a_i, i) \sim (a_j, j) \text{ iff there are } f : A_i \to A_k \text{ and } g : A_j \to A_k \text{ in the diagram such that } f(a_i) = g(a_j).$$

and inclusions $\iota_i : A_i \to K$ are just the standard $a \mapsto (a, i)$.

Proof. First we need to show that $\sim$ is an equivalence relation. For reflexivity, we have an identity morhpism $A_i \to A_i$, so because $\mathcal{I}$ is filtered we have a set A_j with a morphism $A_i \to A_j$, so $(a_i, i) \sim (a_i, i)$. Symmetry is obvious. For transitivity, suppose $(a_i, i) \sim (a_j, j)$, witnessed by some A_{ij}, and $(a_j, j) \sim (a_k, k)$, witnessed by some A_{jk}, as in the solid arrows in the diagram below:

$$\begin{tikzcd}
 & A_j \arrow[d] & & A_j \arrow[d] & \\
A_i \arrow[r] & A_{ij} \arrow[dr, dashed] & & A_{jk} \arrow[dl, dashed] & A_k \arrow[l] \\
 & & A_l \arrow[d, squiggly] & & \\
 & & A'_l & &
\end{tikzcd}$$

Then the dotted arrows and zigzag arrows come from the fact that $\mathcal{I}$ is filtered, so the compositions $A_i \to A'_l$ and $A_k \to A'_l$ show that $(a_i, i) \sim (a_k, k)$. Hence $\sim$ is an equivalence relation.

Now we check that that ι_i commute with the morphisms in the diagram. Suppose $m : j \to k$ is a morphism in $\mathcal{I}$. We need to show that $(a, j) = \iota_j(a) = \iota_k \circ Fm(a) = (Fm(a), k)$ for all $a \in A_j$, which by the equivalence relation amounts to finding some $l \in \mathcal{I}$ and maps $f : A_j \to A_l$ and $g : A_k \to A_l$ such that $f(a) = g(Fm(a))$. Since $\mathcal{I}$ is filtered, there exists some l' with maps $\phi_j : A_j \to A_{l'}$ and $\phi_k : A_k \to A_{l'}$. Now $\phi_{l'}$ and $\phi_k \circ Fm$ are two maps $A_j \to A_{l'}$, so again by filteredness there is some A_l and a map $\phi : A_{l'} \to A_l$ such that $\phi \circ \phi_{l'} = \phi \circ \phi_k \circ Fm$. Then taking $f = \phi \circ \phi_{l'}$ and $g = \phi \circ \phi_k$ works.

Finally for the universal property. Given another set L with maps $\psi_i : A_i \to L$ commuting with the diagram, define $h : K \to L$ via $(a_i, i) \mapsto \psi_i(a_i)$. This commutes with the ι_i trivially, so we just need to check it is well-defined. Suppose $(a_i, i) \sim (a_j, j)$. Then there is some k with maps $f : A_i \to A_k$ and $g : A_j \to A_k$ such that $f(a_i) = g(a_j)$. Since the ψ_i commute with these maps, we get $\psi_i = \psi_k(f(a_i)) = \psi_k(g(a_j)) = \psi_j(a_j)$, completing the proof. □

Exercise (1.3.F)**.** *Let $\mathcal{I}$ be a filtered index category, $F : \mathcal{I} \to \mathsf{Mod}_A$ a diagram. Define the set K as in the previous exercise with the M_i. Addition in K is defined as follows. Given (m_i, i) and (m_j, j), choose*

$l \in \mathcal{I}$ with arrows $u : i \to l$ and $v : j \to l$, and define $(m_i, i) + (m_j, j)$ to be $(Fu(m_i) + Fv(m_j), l)$. Show that K is the colimit of F.

Proof. Let us show that K is an A-module, upon which the previous exercise shows it is the colimit. To see that addition is well-defined, we need to show it is independent of a few choices:

(1) Choice of representative for addition. Suppose $(m_i, i) \sim (m_{i'}, i')$ via maps $\alpha : i \to k$ and $\alpha' : i' \to k$. Choose some l' with $u' : i' \to l'$ and $v' : j \to l'$. Since $\mathcal{I}$ is filtered, there is some $m \in \mathcal{I}$ receiving maps from l, l', k, say $\phi_{lm}, \phi_{l'm}, \phi_{km}$. Now we can (co)equalize $F\phi_{lm} \circ Fv$ and $F\phi_{l'}m \circ Fv'$ via $F\alpha_1$. Then equalize $F(\alpha_1 \circ F\phi_{lm} \circ u)$ and $F(\alpha_1 \circ \phi_{km} \circ \alpha)$ via $F\alpha_2$. Finally, equalize $F(\alpha_2 \circ \alpha_1 \circ \phi_{km} \circ \alpha')$ and $F(\alpha_2 \circ \alpha_1 \circ \phi_{l'm} \circ u')$ via α_3. Then the image of $Fu(m_i) + Fv(m_j)$ under $F(\alpha_3 \circ \alpha_2 \circ \alpha_1 \circ \phi_{lm})$ is equal to the the image of $F\phi_{lm}(Fv(m_j)) + F(\phi_{km}(alpha))(m_i)$ under $F(\alpha_3 \circ \alpha_2 \circ \alpha_1)$, which is equal to the image of $F\phi_{l'm}(Fv'(m_j)) + F(\phi_{km}(alpha'))(m_i')$, which is equal to the image of $Fu'(m_i') + Fv'(m_j)$ under $F(\alpha_3 \circ \alpha_2 \circ \alpha_1 \circ \phi_{l'm})$. This shows that $(Fu'(m_i') + Fv'(m_j), l) \sim (Fu(m_i) + Fv(m_j), l)$. This can be pictured using the diagram below, but beware because the two squares at the top do not necessarily commute:

$$\begin{tikzcd}
M_j \arrow[dr, "Fv"] & & M_i \arrow[dl, "Fu"'] \arrow[dr, "F\alpha"] & & M_{i'} \arrow[dl, "F\alpha'"'] \arrow[dr, "Fu'"] & & M_j \arrow[dl, "Fv'"'] \\
& M_l \arrow[drr, "F\phi_{lm}"'] & & M_k \arrow[d, "F\phi_{km}"'] & & M_{l'} \arrow[dll, "F\phi_{l'm}"] & \\
& & & M_m \arrow[d, "F\alpha_1"'] & & & \\
& & & M_{m'} \arrow[d, "F\alpha_2"'] & \text{equalize } M_j \rightrightarrows M_m & & \\
& & & M_{m''} \arrow[d, "F\alpha_3"'] & \text{equalize } M_i \rightrightarrows M_{m'} & & \\
& & & M_{m'''} & \text{equalize } M_{i'} \rightrightarrows M_{m''} & &
\end{tikzcd}$$

(2) Choice of l. Suppose we take some other $l' \in \mathcal{I}$ with arrows $u' : i \to l'$ and $v' : j \to l'$. Then there is some m with arrows $w : l \to m$ and $w' : l' \to m$. Equalize $w \circ u$ and $w' \circ u'$ via $\alpha_1 : m \to m'$, then equalize $\alpha_1 \circ m \circ v$ and $\alpha_1 \circ m' \circ v'$ via $\alpha_2 : m' \to m''$. Then $F(\alpha_2 \circ \alpha_1 \circ w \circ u)(m_i) + F(\alpha_2 \circ \alpha_1 \circ w \circ v)(m_j) = F(\alpha_2 \circ \alpha_1 \circ w' \circ u')(m_i) + F(\alpha_2 \circ \alpha_1 \circ w' \circ v)(m_j)$, so the maps $F(\alpha_2 \circ \alpha_1 \circ w)$ and $F(\alpha_2 \circ \alpha_1 \circ w')$ give the equivalence between $Fu(m_i) + Fv(m_j)$ and $Fu'(m_i) + Fv'(m_j)$.

(3) Choice of u and v. Suppose we have another morphism $u' : i \to l$. We can equalize them via some map $\alpha : l \to k$. Then it follows that $F\alpha(Fu(m_i) + Fv(m_j)) = F\alpha(Fu'(m_i) + Fv(m_j))$.

Finally we check well-definedness of scalar multiplication. Let $a \in A$ and $(m_i, i) \in K$. Define $a \cdot (m_i, i) := (a \cdot m_i, i)$. To see this is well-defined, take some other $(m_j, j) \sim (m_i, i)$, with morphisms u, v as above. Then $Fu(a \cdot m_i) = a \cdot Fu(m_i) = a \cdot Fv(m_j) = Fv(a \cdot m_j)$ by virtue of Fu being a morphism of A-modules, so scalar multiplication is well-defined. □

Exercise (1.3.G). *Suppose S is a multiplicative subset of A. Interpret the statement $S^{-1}A = \operatorname{colim} \frac{1}{s}A$.*

Solution. This is like 1.3.D.a. Regard S as a category whose objects are S, with a morphism $s \to s'$ if there exists $t \in S$ such that $ts = s'$. Then our diagram will be the functor $s \mapsto \frac{1}{s}A$, which sends a morphism $s \to s'$ to the morphism $\phi_{ss'} : a/s \mapsto at/s'$. We claim that $S^{-1}A$ is the colimit of this.

The maps $\iota_s : \frac{1}{s}A \to S^{-1}A$ are given by inclusion. These commute with the morphisms with S since $\iota_s(a/s) = a/s = at/s' = \iota_{s'}(\phi_{ss'}(a/s))$. Given another A-module K with morphisms $g_s : \frac{1}{s}A \to K$, we define a map $\phi : S^{-1}A \to K$ by $a/s \mapsto g_s(a/s)$. This is well defined by the same logic as 1.3.D.a and commutes with the ι_s since $\phi(\iota_s(a/s)) = g_s(a/s)$. □

Exercise (1.3.H)**.** *Let $F : \mathcal{I} \to \mathsf{Mod}_A$ be a diagram (write M_i for Fi). Show that the colimit of F is given by $K := \oplus_{i\in\mathcal{I}} M_i$ modulo the relations $m_i - F(n)(m_i)$ for each arrow $n : i \to j$ in $\mathcal{I}$.*

Proof. The inclusions $\iota_i : M_i \to K$ send m_i to the element in the direct sum which is m_i in index i and zero everywhere else. Then they commute with the diagram by definition of K. Given a family of maps $g_i : M_i \to L$ for some other A-module L, commuting with the diagram, we can define a map $\phi : K \to L$ via $(m_i)_{i\in\mathcal{I}} \mapsto \sum_{i\in\mathcal{I}} g_i(m_i)$. This is well-defined since $\phi(\iota_i(m_i)) = g_i(m_i) = g_j(Fn(m_i)) = \phi(\iota_j(Fn(m_i)))$, and the same thing shows that it commutes with the families of maps ι_i and g_i. □

1.4. **Adjoints.**

Exercise (1.4.A)**.** *Write down the naturality square in the second argument for an adjoint pair $F : \mathcal{A} \rightleftarrows \mathcal{B} : G$ with natural transformation $\tau_{AB} : \mathrm{Hom}_{\mathcal{B}}(FA, B) \to \mathrm{Hom}_{\mathcal{A}}(A, GB)$.*

Solution. For each $g : B \to B'$ in $\mathcal{B}$, we require that
$$\begin{tikzcd} \mathrm{Hom}_{\mathcal{A}}(FA,B) \arrow[r, "g_*"] \arrow[d, "\tau_{AB}"'] & \mathrm{Hom}_{\mathcal{A}}(FA,B') \arrow[d, "\tau_{AB'}"] \\ \mathrm{Hom}_{\mathcal{B}}(A,GB) \arrow[r, "Gg_*"'] & \mathrm{Hom}_{\mathcal{B}}(A,GB') \end{tikzcd}$$
commutes.

Here g_* is postcomposition by g, and Gg_* is postcomposition by Gg. □

Exercise (1.4.B)**.** *Show that for an adjoint pair as above, the following properties are satisfied.*

(1) *For each A there is a map $\eta : A \to GFA$ (called the unit) such that for any $g : FA \to B$, the corresponding $\tau_{AB}g : A \to GB$ is given by the composition $A \xrightarrow{\eta} GFA \xrightarrow{Gg} GB$.*
Proof. Let H be the functor $\mathrm{Hom}_{\mathcal{A}}(A, G(-)) : \mathcal{B} \to \mathsf{Set}$. By the Yoneda lemma, natural transformations
$$\tau_{A,-} : \mathrm{Hom}_{\mathcal{B}}(FA, -) \Rightarrow H$$
are in bijection with elements of $H(FA) = \mathrm{Hom}_{\mathcal{A}}(A, GFA)$. Call this bijection Ψ. We claim that $\Psi(\tau_{A,-})$ is the desired η. To see this, we know that under the inverse of this bijection, $\Psi(\tau_{A,-})$ maps to the function
$$\mathrm{Hom}_{\mathcal{B}}(FA, B) \to HB, \quad g \mapsto H(g)(\Psi(\tau_{A,B})) = Gg \circ \Psi(\tau_{A,B})$$
but because Ψ is a bijection this function must coincide with $\tau_{A,B}$, so $\tau_{A,B} = Gg\circ\Psi(\tau_{A,B})$. □

(2) *For each B there is a map $\epsilon : FG(B) \to B$ (called the counit) so that for each $f : A \to GB$, the corresponding $\tau^{-1}_{AB}(f) : FA \to B$ is given by the composition $F(A) \xrightarrow{f} FGB \xrightarrow{\epsilon} B$.*
Proof. Let H be the functor $\mathrm{Hom}_{\mathcal{B}}(F(-), B) : \mathcal{A} \to \mathsf{Set}$. By the contravariant Yoneda lemma, natural transformations
$$\tau^{-1}_{-,B} : \mathrm{Hom}_{\mathcal{A}}(-, GB) \to H$$
are in bijection with elements of $H(GB) = \mathrm{Hom}_{\mathcal{B}}(FGB, B)$, call this bijection Ψ. Then $\Psi : \tau^{-1}_{AB}$ is the desired ϵ. To see this, under the inverse of Ψ we know that $\Psi(\tau^{-1}_{-,B})$ maps to the function
$$\mathrm{Hom}(A, GB) \to \mathrm{Hom}(FA, B), f \mapsto Hf(\Psi(\tau^{-1}_{A,B})) = \Psi(\tau^{-1}_{A,B}) \circ Ff$$
so $\tau^{-1}_{-,B}(f) = \Psi(\tau^{-1}_{A,B}) \circ Ff$. □

Exercise (1.4.C)**.** *Let M, N, P be A-modules. Describe a bijection between* $\mathrm{Hom}(M \otimes_A N, P)$ *and* $\mathrm{Hom}(M, \mathrm{Hom}_A(N, P))$.

Solution. Let us start with $\Phi_{M,P} : \mathrm{Hom}(M \otimes_A N, P) \to \mathrm{Hom}(M, \mathrm{Hom}_A(N, P))$. By the universal property of tensor products, the data of $f \in \mathrm{Hom}(M \otimes N, P)$ is equivalent to the data of a bilinear map $\tilde{f} : M \times N \to P$ satisfying $f(m \otimes n) = \tilde{f}(m, n)$. So define $\Phi_{M,P}(f)$ via $\Phi_{M,P}(f)(m) = \tilde{f}(m, -)$.

In the other direction, define $\Psi_{M,P} : \mathrm{Hom}(M, \mathrm{Hom}_A(N, P)) \to \mathrm{Hom}(M \otimes_A N, P)$ as follows. Given g in the domain, $g(m)$ is an A-linear map $N \to P$, so $g(-)(-) : M \times N \to P$ is bilinear. Hence it induces a map $M \otimes N \to P$ which we call $\Psi_{M,P}(g)$, satisfying $\Psi_{M,P}(g)(m \otimes n) = g(m)(n)$.

To see these are mutually inverse, we check that

$$\Psi_{M,P}(\Phi_{M,P}(f))(m \otimes n) = \Phi_{M,P}(f)(m)(n) = \tilde{f}(m, n) = f(m \otimes n).$$

$$\Phi_{M,P}(\Psi_{M,P}(g))(m)(n) = \widetilde{\Psi_{M,P}}(g)(m, n) = \Psi_{M,P}(g)(m \otimes n) = g(m)(n)$$

which completes the bijection. □

Exercise (1.4.D)**.** *Show that* $(-) \otimes N$ *and* $\mathrm{Hom}(N, -)$ *are adjoint.*

Proof. Let us stick to the notation above. We need to show that Φ is natural in M and P. So let $\pi : M \to M'$ and $\tau : P \to P'$ be morphisms. In the first argument, we have the diagram

$$\begin{tikzcd} \mathrm{Hom}(M' \otimes N, P) \arrow[r, "\Phi_{M',P}"] \arrow[d, "\pi^*"'] & \mathrm{Hom}(M', \mathrm{Hom}(N, P)) \arrow[d, "\circ\pi"] \\ \mathrm{Hom}(M \otimes N, P) \arrow[r, "\Phi_{M,P}"'] & \mathrm{Hom}(M, \mathrm{Hom}(N, P)) \end{tikzcd}$$

We verify for f in the upper left corner that

$$\Phi_{M',P}(f)(\pi(m))(n) = \tilde{f}(\pi(m), n) = f(\pi(m) \otimes n)$$

and

$$\Phi_{M,P}(f \circ \pi_*)(m)(n) = \widetilde{f \circ \pi^*}(n, m) = f(\pi^*(n \otimes m)) = f(\pi(n) \otimes m).$$

For the other diagram:

$$\begin{tikzcd} \mathrm{Hom}(M \otimes N, P) \arrow[r, "\Phi_{M,P}"] \arrow[d, "\tau\circ-"'] & \mathrm{Hom}(M, \mathrm{Hom}(N, P)) \arrow[d, "\tau_*"'] \\ \mathrm{Hom}(M \otimes N, P') \arrow[r, "\Phi_{M,P'}"] & \mathrm{Hom}(M, \mathrm{Hom}(N, P')) \end{tikzcd}$$

Take g in the upper left corner, and verify:

$$\tau_*(\Phi_{M,P}(g))(m)(n) = \tau \circ \Phi_{M,P}(g)(m)(n) = \tau(g(m \otimes m))$$

and

$$\Phi_{M,P'}(\tau \circ g)(m)(n) = \tau(g(m \otimes m))$$

which completes the proof. □

Exercise (1.4.E)**.** *Let $\phi : B \to A$ be a morphism of rings. For an A-module M, we can regard it as a B-module M_B via this morphism, which yields a functor $(-)_B : \mathsf{Mod}_A \to \mathsf{Mod}_B$. Show that this functor is right adjoint to $(-) \otimes_B A$.*

Proof. Let us describe the bijection $\Phi : \mathrm{Hom}(N \otimes_B A, M) \to \mathrm{Hom}(N, M_B)$. Start with f on the left. Define $\Psi(f)(n) := f(n \otimes 1_A)$. This is a valid B-module morphism, since $\Psi(f)(b \cdot n) = f(b \cdot n \otimes 1_A) = f(n \otimes \phi(b)) = \phi(b) \cdot f(n \otimes 1) = b \cdot f(n \otimes 1)$.

On the other hand, given g on the right, first consider the map $\tilde{\Psi}(g) : N \times A \to M$ given by $(n, a) \mapsto a \cdot g(n)$. This is B-bilinear since $\Psi(g)(b \cdot n, a) = a \cdot g(b \cdot n) = a \cdot \phi(b) \cdot g(n) = \phi(b) \cdot (a \cdot g(n)) = \phi(b) \cdot \Psi(g)(n, a)$, and in the second variable $\Psi(g)(n, \phi(b)a) = \phi(b)a \cdot g(n) = \phi(b) \cdot (a \cdot g(n))$. Hence $\tilde{\Psi}(g)$ factors through the tensor product to yield a map $\Psi(g) : N \otimes A \to M$ with $\Psi(g)(n \otimes a) = \Psi(\tilde{g})(n, a) = a \cdot g(n)$.

These are mutually inverse since

$$\Phi(\Psi(g))(n) = \Psi(g)(n \otimes 1) = 1 \cdot g(n) = g(n)$$

and

$$\Psi(\Phi(f))(n \otimes a) = a \cdot \Phi(f)(n) = a \cdot f(n \otimes 1) = f(n \otimes a).$$

For naturality, let $\pi : N \to N'$ and $\tau : M \to M'$ be morphisms. The first square to check is

$$\begin{tikzcd}
\mathrm{Hom}_A(N' \otimes A, M) \arrow[r, "\Phi_{N',M}"] \arrow[d, "\pi^*"'] & \mathrm{Hom}_B(N', M_B) \arrow[d, "\circ\pi"'] \\
\mathrm{Hom}_A(N \otimes_B A, M) \arrow[r, "\Phi_{N,M}"'] & \mathrm{Hom}_B(N, M_B)
\end{tikzcd}$$

Choose f in the top left, then we get

$$\Phi_{N',M}(f)(\pi(n)) = f(\pi(n) \otimes 1)$$

$$\Phi_{N,M}(\pi^*(f))(n) = \pi^* f(n \otimes 1) = f(\pi(n) \otimes 1).$$

The other square is

$$\begin{tikzcd}
\mathrm{Hom}_A(N \otimes A, M) \arrow[r, "\Phi_{N,M}"] \arrow[d, "\tau\circ"'] & \mathrm{Hom}(N, M_B) \arrow[d, "\tau\circ"] \\
\mathrm{Hom}_A(N \otimes A', M) \arrow[r, "\Phi_{N,M'}"'] & \mathrm{Hom}(N, M'_B)
\end{tikzcd}$$

Choosing g in the top left we get

$$\tau \circ \Phi_{N,M}(g)(n) = \tau(g(n \otimes 1))$$

$$\Phi_{N,M'}(\tau \circ g) = \tau(g(n \otimes 1))$$

which proves the adjunction. □

Exercise (1.4.F)**.** *Show that if an abelian semigroup S is already an abelian group then its groupification is the identity morphism $S \to S$.*

Proof. Let G be an abelian group and let $f : S \to G$ be a map of abelian semigroups. Then clearly $f \circ \mathrm{id} = f$, so f factors through the identity map uniquely. It remains to check that f is an abelian group homomorphism. We have

$$f(0) = f(0 + 0) = f(0) + f(0)$$

so $f(0) = 0$ and

$$f(0) = f(a - a) = f(a) + f(-a)$$

so $f(-a) = -f(a)$. □

Exercise (1.4.G). *Construct the groupification functor H from the category of nonempty abelian semigroups to the category of groups, and show it is left adjoint to the forgetful functor.*

Proof. Define $H(S)$ to be the set $S \times S$ modulo the equivalence relation $(a,b) \sim (c,d)$ if there exists $e \in S$ such that $a+d+e = b+c+e$. Let's check this is an equivalence relation. Clearly $(a,b) \sim (a,b)$ since we can take e to be any element. Symmetry is also obvious. For transitivity, suppose $a+d+e = b+c+e$ and $c+y+e' = d+x+e'$. Then adding both equations shows that $(a,b) \sim (x,y)$. The group operation is given by $(a,b)+(c,d) = (a+c, b+d)$. The identity is (a,a) for any $a \in S$, and the inverse of (a,b) is (b,a), so $H(S)$ is indeed an abelian group. Define the map $\phi : S \to H(S)$ by $s \mapsto (s+s, s)$. This is an abelian semigroup morphism since $s+t \mapsto (s+t+s+t, s+t) = (s+s,s)+(t+t,t)$.

Now let $f : S \to G$ be a map of abelian semigroups. Define $\tilde{f} : H(S) \to G$ via $\tilde{f}(a,b) = f(a) - f(b)$. This is well-defined: suppose $a+d+e = b+c+e$. Then $\tilde{f}(c,d) = f(c) - f(d) = f(a) - f(b) + f(e) - f(e) = \tilde{f}(a,b)$. And f we have $f(s) = \tilde{f}(\phi(s)) = \tilde{f}(s+s,s) = f(s+s) - f(s) = f(s)$. Suppose we have another map g with $f = g \circ \phi$. Then g and $\tilde{f}$ agree on elements of the form $(s+s,s)$. Since we can rewrite any $(s,t) \in H(S)$ as $(s+s,s)-(t+t,t)$, we get that $g(s,t) = g(s+s,s) - g(t+t,t) = g(\phi(s)) - g(\phi(t)) = \tilde{f}(s,t)$. Given a map $\pi : S \to S'$, we define $H(\pi) : H(S) \to H(S')$ via $(s,t) \mapsto (\pi(s), \pi(t))$, making H into a functor.

Now to show adjointness. Let F be the forgetful functor and G and abelian group. Consider the function $\Phi : \mathrm{Hom}(H(S), G) \to \mathrm{Hom}(S, FG)$ given by $\Phi(f)(s) = f(s+s, s)$. In the other direction define $\Psi : \mathrm{Hom}(S, FG) \to \mathrm{Hom}(H(S), G)$ by $\Psi(g)(a,b) = g(a) - g(b)$, which is well-defined by the same reason as above. We check that

$$\Phi(\Psi(g))(s) = \Psi(g)(s+s,s) = g(s+s) - g(s) = g(s)$$
$$\Psi(\Phi(f))(a,b) = \Phi(f)(a) - \Phi(f)(b) = f(a+a,a) - f(b+b,b) = f(a,b).$$

Now for naturality, let $\pi : S \to S'$ be a map of abelian semigroups and $\tau : G \to G'$ an abelian group homomorphism. The first square to check is

$$\begin{tikzcd} \mathrm{Hom}(H(S'),G) \arrow[r, "\Phi"] \arrow[d, "-\circ H\pi"'] & \mathrm{Hom}(S',FG) \arrow[d, "-\circ\pi"] \\ \mathrm{Hom}(H(S),G) \arrow[r, "\Phi"'] & \mathrm{Hom}(S,FG) \end{tikzcd}$$

where $\Phi(f)(\pi(s)) = f(\pi(s)+\pi(s), \pi(s))$ and $\Phi(f \circ H\pi)(s) = (f \circ H\pi)(s+s,s) = f(\pi(s)+\pi(s), \pi(s))$. The second square to check is

$$\begin{tikzcd} \mathrm{Hom}(H(S),G) \arrow[r, "\Phi"] \arrow[d, "\tau\circ-"'] & \mathrm{Hom}(S,FG) \arrow[d, "F\tau\circ-"] \\ \mathrm{Hom}(H(S),G') \arrow[r, "\Phi"'] & \mathrm{Hom}(S,FG') \end{tikzcd}$$

where $F\tau(\Phi(g))(s) = \tau(g(s+s,s))$ and $\Phi(\tau \circ g)(s) = \tau(g(s+s,s))$, completing the proof. □

Exercise (1.4.H). *Let S be a multiplicative subset of A. Show that $S^{-1}(-) : \mathsf{Mod}_A \to \mathsf{Mod}_{S^{-1}A}$ is left adjoint to the forgetful functor $F : \mathsf{Mod}_{S^{-1}A} \to \mathsf{Mod}_A$.*

Proof. Let us first construct the bijection for $M \in \mathsf{Mod}_A$ and $N \in \mathsf{Mod}_{S^{-1}A}$

$$\Phi : \mathrm{Hom}_{\mathsf{Mod}_{S^{-1}A}}(S^{-1}M, N) \to \mathrm{Hom}_{\mathsf{Mod}_A}(M, FN).$$

by sending f on the left to $\Phi(f)(m) := f(m/1)$. In the other direction send g on the right to $\Psi(g)(n/s) = g(n)/s$. These are clearly compatible with the module actions. Then

$$\Phi(\Psi(g))(m) = \Psi(g)(m/1) = g(m)/1 = g(m)$$

$$\Psi(\Phi(f))(n/s) = \Phi(f)(n)/s = f(n/1)/s = f(n/s).$$

Let's check the squares! For morphisms $\pi : M \to M'$ and $\tau : N \to N'$, we have

$$\begin{tikzcd} \mathrm{Hom}_{\mathsf{Mod}_{S^{-1}A}}(S^{-1}M', N) \arrow[r, "\Phi"] \arrow[d, "-\circ S^{-1}\pi"'] & \mathrm{Hom}_{\mathsf{Mod}_A}(M', FN) \arrow[d, "-\circ\pi"] \\ \mathrm{Hom}_{\mathsf{Mod}_{S^{-1}A}}(S^{-1}M, N) \arrow[r, "\Phi"'] & \mathrm{Hom}_{\mathsf{Mod}_A}(M, FN) \end{tikzcd}$$

where $\Phi(f)(\pi(m)) = f(\pi(m)/1)$ and $\Phi(f \circ S^{-1}\pi)(m) = \Phi(f)(S^{-1}\pi(m)) = f(\pi(m)/1)$. The other square is

$$\begin{tikzcd} \mathrm{Hom}_{\mathsf{Mod}_{S^{-1}A}}(S^{-1}M, N) \arrow[r, "\Phi"] \arrow[d, "\tau\circ-"'] & \mathrm{Hom}_{\mathsf{Mod}_A}(M, FN) \arrow[d, "F\tau\circ-"] \\ \mathrm{Hom}_{\mathsf{Mod}_{S^{-1}A}}(S^{-1}M, N') \arrow[r, "\Phi"'] & \mathrm{Hom}_{\mathsf{Mod}_A}(M, FN') \end{tikzcd}$$

where $F\tau \circ \Phi(g)(m) = \tau(g(m/1))$ and $\Phi(\tau \circ g)(m) = \tau(g(m/1))$ which completes the proof. □

1.5. An introduction to abelian categories.

Exercise (1.5.H)**.** *Given a complex $A^\bullet$, describe exact sequences*

$$0 \to \operatorname{im} f^i \to A^{i+1} \to \operatorname{coker} f^i \to 0$$
$$0 \to H^i(A) \to \operatorname{coker} f^{i-1} \to \operatorname{im} f^i \to 0$$

Solution. Let us first show that kernels are monomorphisms and cokernels are epimorphisms. Let $\phi : A \to B$ be a morphism, and let $u : \ker\phi \to A$, $v : B \to \operatorname{coker}\phi$. Suppose we have morphisms $f, g : K \to \ker\phi$ such that $u \circ f = u \circ g$. Then we have the following commutative diagram,

$$\begin{tikzcd} K \arrow[dr, shift left, "f"] \arrow[dr, shift right, "g"'] \arrow[drr, bend left] \arrow[ddr, bend right, "u\circ f = u\circ g"'] & & \\ & \ker\phi \arrow[r] \arrow[d, "u"] & 0 \arrow[d] \\ & A \arrow[r, "\phi"'] & B \end{tikzcd}$$

which implies $f = g$. Likewise, suppose we have morphisms $f', g' : \operatorname{coker} \to L$ such that $f' \circ v = g' \circ v$. Then we get the following commutative diagram

$$\begin{tikzcd} A \arrow[r] \arrow[d, "\phi"'] & 0 \arrow[d] \arrow[ddr, bend left] & \\ B \arrow[r, "v"'] \arrow[drr, bend right, "f'\circ v = g'\circ v"'] & \operatorname{coker}\phi \arrow[dr, shift left, "g'"] \arrow[dr, shift right, "f'"'] & \\ & & L \end{tikzcd}$$

which implies $f' = g'$.

Onto the problem. The first sequence can be written as

$$0 \to \ker(\operatorname{coker} f^i) \xrightarrow{\alpha} A^{i+1} \xrightarrow{\beta} \operatorname{coker} f^i \to 0.$$

By what we have just done, this is exact in the first and third entry. Because in abelian categories every monomorphism is the kernel of its cokernel, we have

$$\operatorname{im}\alpha = \ker\operatorname{coker}\alpha = \alpha = \ker\operatorname{coker} f^i = \ker\beta$$

here we are switching back and forth between referring to the map in the (co)kernel and the objects involved. This shows exactness in the middle.

For the next sequence... epic fail when I tried to do it in an arbitrary abelian category, so let us just stick to Vakil's suggestion of working with modules for now. We can rewrite the sequence as

$$0 \to \ker f^i / \operatorname{im} f^{i-1} \xrightarrow{\alpha} A^i / \operatorname{im} f^{i-1} \xrightarrow{\beta} \operatorname{im} f^i \to 0.$$

Then we can define α to be the map induced by the composition $\ker f^i \to A^i \to A^i / \operatorname{im} f^i$, which obviously factors through $\operatorname{im} f^{i-1}$. This is injective, making the sequence exact in the first entry. For β, let it be the map induced by f^i, which factors through $\operatorname{im} f^{i-1}$ since A is a complex. This is again surjective since it is just a quotient. We have $\beta(\alpha([x])) = f(x) = 0$ so $\operatorname{im} \alpha \subset \ker \beta$, and conversely if $\beta([y]) = 0$ then $y \in \ker f$ which completes the proof. □

Exercise (1.5.B)**.** *Let*

$$0 \xrightarrow{d^0} A^1 \xrightarrow{d^1} \cdots \xrightarrow{d^{n-1}} A^n \xrightarrow{d^n} 0$$

be a cochain complex of finite dimensional vector spaces. Define $h^i(A) = \dim H^i(A)$*. Show that* $\sum(-1)^i \dim A^i = \sum(-1)^i h^i(A)$*.*

Proof. Let us begin with the right. We know

$$h^i(A) = \dim H^i(A) = \dim(\ker d^i / \operatorname{im} d^{i-1}) = \dim\ker d^i - \dim\operatorname{im} d^{i-1}$$

so taking the alternating sum we have

$$\begin{aligned}\sum(-1)^i h^i(A) &= \sum(-1)^i(\dim\ker d^i - \dim\operatorname{im} d^{i-1}) \\ &= (\dim\ker d^0 - \dim\operatorname{im} d^{-1}) \\ &- \dim\ker d^1 + \dim\operatorname{im} d^0 \\ &+ \cdots \\ &+ (-1)^n(\dim\ker d^n - \dim\operatorname{im} d^{n-1})\end{aligned}$$

but note that $\dim\ker d^n = \dim A^n$ and $\dim\operatorname{im} d^{-1} = 0$, so by rank-nullity this sum, after regrouping, is equal to $\sum(-1)^i \dim A^i$ as desired. □

Exercise (1.5.C)**.** *Let $\mathcal{C}$ be an abelian category. Show that the category of complexes in $\mathcal{C}$, denoted* $\mathsf{Ch}(\mathcal{C})$*, is also abelian.*

Proof. Let us show the axioms one by one.

A1 *Homsets are abelian groups.* Let $A, B \in \mathsf{Ch}(\mathcal{C})$. Then $\operatorname{Hom}_{\mathsf{Ch}(\mathcal{C})}(A, B)$ is the subgroup of $\prod_{n\in\mathbf{Z}} \operatorname{Hom}_{\mathcal{C}}(A^n, B^n)$ consisting of those morphisms that commute with the maps in A and B. This distributes over addition by virtue of the individual morphisms distributing in each A^n and B^n.

A2 *Zero object* The zero object is given by $\cdots \to 0 \to 0 \to 0 \to \cdots$, where 0 is the zero object in $\mathcal{C}$.

A3 *Finite products.* Let $A, B \in \mathsf{Ch}(\mathcal{C})$. We claim that their product is given by the chain complex whose nth entry is $A^n \times B^n$, and differential defined as follows. Take $A^n \times B^n$ with the projections π_{A^n} and π_{B^n}. Postcompose these with the differentials $A^n \to A^{n+1}$ and $B^n \to B^{n+1}$. This gives maps from $A^n \times B^n \to A^{n+1}$ and $A^n \times B^n \to B^{n+1}$, which by the universal property of products yields our map $d^n_{A\times B} : A^n \times B^n \to A^{n+1} \times B^{n+1}$. Then $d^2_{A\times B} = 0$ by commutativity of the following diagram:

$$\begin{tikzcd}
& A^n \arrow[rr, "d_A^n"] & & A^{n+1} \arrow[rr, "d_A^{n+1}"] & & A^{n+2} \\
A^n \times B^n \arrow[ur] \arrow[dr] \arrow[rr, dashed, "d_{A\times B}^n"] & & A^{n+1} \times B^{n+1} \arrow[ur] \arrow[dr] \arrow[rr, "d_{A\times B}^{n+1}"] & & A^{n+2} \times B^{n+2} \arrow[ur] \arrow[dr] & \\
& B^n \arrow[rr, "d_B^n"'] & & B^{n+1} \arrow[rr, "d_B^{n+1}"'] & & B^{n+2}
\end{tikzcd}$$

To show this satisfies the universal property of the product, suppose we have maps $f : C \to A$ and $g : C \to B$ in $\mathsf{Ch}(\mathcal{C})$. Then by definition we get maps $f^n : C^n \to A^n$ and $g^n : C^n \to B^n$ which by the universal property of products in $\mathcal{C}$, we get a map $h^n : C^n \to A^n \times B^n$. Then the h^n assemble into a map $C \to A \times B$ as desired.

A4 *Every map has a kernel and a cokernel.* Let $f^\bullet : A^\bullet \to B^\bullet$ be a map of complexes. We claim that its kernel is given by the complex $K^\bullet$ with $K^n = \ker f^n$ and levelwise maps $\iota^n : K^n \to A^n$. We first need to elaborate on what the differentials in K are. Consider the following diagram:

$$\begin{tikzcd}
\ker f^n \arrow[r, dashed] \arrow[d, "\iota^n"'] \arrow[dr, "d_A^n \circ \iota^n"] & \ker f^{n+1} \arrow[d, "\iota^{n+1}"] \\
A^n \arrow[r, "d_A^n"'] \arrow[d, "f^n"'] & A^{n+1} \arrow[d, "f^{n+1}"] \\
B^n \arrow[r, "d_B^n"'] & B^{n+1}
\end{tikzcd}$$

Since the bottom square commutes, the diagonal map composed with f^{n+1} is zero, so $d_A^n \circ \iota^n$ factors through ι^{n+1} to yield our map d_K^n from $\ker f^n$ to $\ker f^{n+1}$. To see $d_K^2 = 0$, we can extend the diagram

$$\begin{tikzcd}
\ker f^n \arrow[r, dashed] \arrow[d, "\iota^n"'] \arrow[dr] & \ker f^{n+1} \arrow[r, dashed] \arrow[d, "\iota^{n+1}"] \arrow[dr] & \ker f^{n+2} \arrow[d, "\iota^{n+2}"] \\
A^n \arrow[r, "d_A^n"'] & A^{n+1} \arrow[r, "d_A^{n+1}"'] & A^{n+1}
\end{tikzcd}$$

which tells us $\iota^{n+2} \circ d_K^2 = d_A^2 \circ \iota^n = 0$, so because ι is a monomorphism we are done. Thus K is a complex. Now we need to check that
$\begin{tikzcd} K \arrow[r] \arrow[d] & 0 \arrow[d] \\ A \arrow[r] & B \end{tikzcd}$
is a Cartesian square. It obviously commutes. Let $\pi : D \to A$ be a map of complexes whose composition with f is zero. Then $f^n \circ \pi^n = 0$ for all n, so there is an induced map $\tilde{\pi}^n : D^n \to K^n$ for all n such that $\iota^n \circ \tilde{\pi}^n = \pi^n$. It follows that $\iota \circ \tilde{\pi} = \pi$, so once we show that $\tilde{\pi}$ is a map of complexes are are done. Consider below:

$$\begin{tikzcd}
D^n \arrow[r, "d_D"] \arrow[d, "\tilde{\pi}^n"'] & D^{n+1} \arrow[d, "\tilde{\pi}^{n+1}"] \\
K^n \arrow[r, "d_K"] & K^{n+1}
\end{tikzcd}
\qquad
\begin{tikzcd}
D^n \arrow[dr, dashed] \arrow[drr, bend left] \arrow[ddr, bend right, "d_A^n \circ \pi^n"'] & & \\
& K^{n+1} \arrow[r] \arrow[d, "\iota^{n+1}"'] & 0 \arrow[d] \\
& A^{n+1} \arrow[r] & B^{n+1}
\end{tikzcd}$$

To see that the guy on the left commutes, we will show that they make the pullback diagram on the right commute if we put them in place of the dotted arrow, whence by the universal property they are equal. We compute for the top-right path

$$\iota^{n+1} \circ \tilde{\pi}^{n+1} \circ d_D = \pi^{n+1} \circ d_D^n = d_A^n \circ \pi^n$$

since π is a map of complexes. And we compute for the bottom-left path

$$\iota^{n+1} \circ d_K^n \circ \tilde{\pi}^n = d_A^n \circ \iota^n \circ \tilde{\pi}^n = d_A^n \circ \pi^n$$

as desired, using the fact that ι is a map of complexes. This completes the proof. The cokernel is exactly the same.

A5 *Every monomorphism is the kernel of its cokernel.* Let $f : A \to B$ be a monomorphism and $p : B \to \operatorname{coker} f$ its cokernel. Since f is a monomorphism, its kernel is zero, so $\ker f^n = 0$, so each f^n is a monomorphism. Now $\ker \operatorname{coker} f$ is the complex which is levelwise $\ker \operatorname{coker} f^n$, and because $\mathcal{C}$ is abelian, this is equal to f^n since it is a monomorphism. Hence the kernel of the cokernel of f is the complex which is levelwise f^n, so it is itself.

A6 *Every epimorphism is the cokernel of its kernel.* By the exact same logic as above. If $g : A \to B$ is an epi, then each g^n is an epi, so $\operatorname{coker} \ker g^n = g^n$, so $\operatorname{coker} \ker g = g$.

Whew! □

Exercise (1.5.D)**.** *Show that a map of complexes $f : A^\bullet \to B^\bullet$ induces a map on cohomology $H^i(A) \to H^i(B)$.*

Proof. Let us work with modules. First note that f^i sends the kernel of d_A^i to the kernel of $d_i(B)$, since $d_B^i(f_i(x)) = f^{i+1} d_A^i(x) = 0$. Moreover, $f^i \circ d_A^{i-1} = d_B^{i-1} \circ f^{i-1}$, so f^i sends the image of d_A^{i-1} to the image of d_B^{i-1}. Hence f^i induces a map $H^i(A) \to H^i(B)$.

Side note: I tried to do this inside an arbitrary abelian category, but could not figure it out... One day I will return. □

Exercise (1.5.E)**.** *Show that two homotopic maps $f, g : C \to D$ induce the same maps on homology.*

Proof. We work with modules again. Let $\omega : C^\bullet \to D^{\bullet-1}$ be the homotopy. Pick some $x \in \ker d_A^i$. We know that

$$(f - g)(x) = d_D^{i-1}(\omega^i(x)) + \omega^{i+1}(d_A^i(x)) = d_D^{i-1}(\omega^i(x)).$$

That is, $f(x)$ and $g(x)$ differ by something in the image of d_D^{i-1}, so they are homologous. □

Exercise (1.5.F)**.** *Show that applying an exact functor $F : \mathcal{C} \to \mathcal{D}$ to an exact sequence preserves exactness.*

Proof. Let $A \xrightarrow{f} B \xrightarrow{g} C$ be exact. We will show that F preserves kernels and images, from which it follows that $\operatorname{im} Ff = F \operatorname{im} f = F \ker g = \ker Fg$.

Let $f : A \to B$ be a morphism. Then by exactness of F and the sequences

$$0 \to \ker f \to A \to B, \quad A \to B \to \operatorname{coker} f \to 0$$

we know that

$$0 \to F(\ker f) \to FA \to FB, \quad FA \to FB \to F(\operatorname{coker} f) \to 0$$

are also exact, so $F(\ker f)$ is a mono and $F(\operatorname{coker} f)$ is an epi. By exactness, we obtain

$$\ker Ff = \operatorname{im}(F(\ker f)) = F \ker f$$

since the image of a mono is itself. And from the second sequence

$$\operatorname{im} Ff = \ker(F \operatorname{coker} f) = F(\ker \operatorname{coker} f) = F(\operatorname{im} f)$$

where in the middle equality we used what we just proved, that F preserves kernels. □

Exercise (1.5.G)**.** *Let A be a ring, S a multiplicative subset, and M an A-module.*

(a) Show that $S^{-1}(-) : \mathsf{Mod}_A \to \mathsf{Mod}_{S^{-1}A}$ is an exact functor.

Proof. Let $N \xrightarrow{f} N' \xrightarrow{g} N''$ be exact. Since $S^{-1}(-)$ is a functor, it is immediate that $S^{-1}g \circ S^{-1}f = 0$, so $\operatorname{im} S^{-1}g \subset \ker S^{-1}f$. To show the other containment, let $a/b \in \ker S^{-1}g$, i.e. $S^{-1}g(a/b) = g(a)/b = 0$. Then there is some $t \in S$ such that $tg(a) = 0$, which gives $g(ta) = 0$, so $ta \in \ker g = \operatorname{im} f$, so let $y \in N$ such that $f(y) = ta$. Then $S^{-1}(y/tb) = ta/tb = a/b$, so $a/b \in \operatorname{im} S^{-1}g$, proving exactness. □

(b) Show that $(-) \otimes M$ is right exact.

Proof. This is Exercise 1.2.H. □

(c) Show that $\operatorname{Hom}(M, -)$ *is left exact. For $\mathcal{C}$ an abelian category and $C \in \mathcal{C}$ show that* $\operatorname{Hom}(C, -)$ *is a left exact functor $\mathcal{C} \to \mathsf{Ab}$.*

Proof. The first part of this follows from the second part. Let $0 \to X \xrightarrow{f} Y \xrightarrow{g} Z$ be exact. Let us first show that f_* is a monomorphism. Let H be an abelian group with morphisms $h_1, h_2 : H \to \operatorname{Hom}(C, X)$ such that $(f_* \circ h_1)(h) = (f_* \circ h_2)(h)$ for all $h \in H$. But f_* is just postcomposition by f, so $f \circ h_1(h) = f \circ h_2(h)$ implies $h_1(h) = h_2(h)$ since f is a mono, so f_* is a mono.

Now for exactness at $\operatorname{Hom}(C, Y)$. Take $\phi \in \operatorname{Hom}(C, X)$. Then $g_* f_*(\phi) = g \circ f \circ \phi$. This is the zero homomorphism since $g \circ f = 0$, hence $\operatorname{im} f_* \subset \ker g_*$. Conversely, take some $\psi \in \ker g_*$, i.e. $g \circ \psi$ is the zero morphism. This implies that ψ factors through $\ker g = \operatorname{im} f = f$ since f is a mono. Hence there exists some ψ' such that $f \circ \psi' = \psi$, so $\psi \in \operatorname{im} f_*$. □

(d) Likewise, show $\operatorname{Hom}(-, M)$ *is left exact. For $\mathcal{C}$ an abelian category and $C \in \mathcal{C}$ show that* $\operatorname{Hom}(-, C)$ *is a left exact contravariant functor $\mathcal{C} \to \mathsf{Ab}$.*

Proof. Let $X \xrightarrow{f} Y \xrightarrow{g} Z \to 0$ be exact. Let us first show that g^* is a monomorphism. Again let H be an abelian group and $h_1, h_2 : H \to \operatorname{Hom}(Z, C)$ such that $(g^* \circ h_1)(h) = (g^* \circ h_2)(h)$ for all $h \in H$. This implies $h_1(h) \circ g = h_2(h) \circ g$, for all h, and because g is an epi, we obtain $h_1 = h_2$, so g^* is a mono.

For exactness in the midle, take $\phi \in \operatorname{Hom}(Z, C)$. Then $f^*(g^*(\phi)) = \phi \circ g \circ f = 0$, so $\operatorname{im} g^* \subset \ker f^*$. Conversely, let $\psi \in \ker f^*$. Then $\psi \circ f = 0$, so ψ factors through $\operatorname{coker} f = \operatorname{im} g$ by exactness. It follows that there exists some $\tilde{\psi} : \operatorname{im} g \to C$ such that $\psi = \tilde{\psi} \circ g$, so $\psi \in \operatorname{im} g^*$. □

Exercise (1.5.H)**.** *Suppose M is a finitely presented A-module, i.e. it fits in an exact sequence*

$$A^{\oplus q} \to A^{\oplus p} \to M \to 0.$$

Use this and the left-exactness of Hom *to describe an isomorphism*

$$S^{-1} \operatorname{Hom}_A(M, N) \to \operatorname{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N).$$

Proof. By the previous exercise, we know that

$$S^{-1}A^{\oplus q} \to S^{-1}A^{\oplus q} \to S^{-1}M \to 0$$

is also exact. Applying $\operatorname{Hom}(-, N)$ to both of these, we get exact sequences

$$0 \to \operatorname{Hom}(M, N) \to \operatorname{Hom}(A^{\oplus p}, N) \to \operatorname{Hom}(A^{\otimes q}, N)$$

$$0 \to \operatorname{Hom}(S^{-1}M, S^{-1}N) \to \operatorname{Hom}(S^{-1}A^{\oplus p}, S^{-1}N) \to \operatorname{Hom}(S^{-1}A^{\otimes q}, S^{-1}N).$$

We know that localization is exact, commutes with direct sum, and maps out of a finite direct sum of modules correspond to a finite direct sum of maps, so the sequences become

$$0 \to S^{-1}\operatorname{Hom}(M,N) \to S^{-1} \oplus_p \operatorname{Hom}(A,N) \to S^{-1} \oplus_q \operatorname{Hom}(A,N)$$

$$0 \to \operatorname{Hom}(S^{-1}M, S^{-1}N) \to \oplus_p \operatorname{Hom}(S^{-1}A, S^{-1}N) \to \oplus_q \operatorname{Hom}(S^{-1}A, S^{-1}N)$$

Finally, for any ring R and R-module L we know $\operatorname{Hom}_R(R,L) \cong L$ naturally, so our exact sequences become

$$0 \to S^{-1}\operatorname{Hom}(M,N) \to (S^{-1}N)^{\oplus p} \to (S^{-1}M)^{\oplus q}$$

and

$$0 \to \operatorname{Hom}(S^{-1}M, S^{-1}N) \to (S^{-1}N)^{\oplus p} \to (S^{-1}M)^{\oplus q}$$

so these two things are the kernel of the same map, hence are isomorphic. □

Exercise (1.5.I)**.** *Let $F : \mathcal{A} \to \mathcal{B}$ be a functor of abelian categories, and let $C^\bullet$ be a complex in $\mathcal{A}$.*

(a) If F is right exact, show that there is a natural morphism $FH^i(C) \to H^i(FC)$.

Proof. Consider the exact sequence $C^i \xrightarrow{d^i} C^{i+1} \to \operatorname{coker} d^i \to 0$. Since F is right exact, the sequence $FC^i \xrightarrow{Fd^i} FC^{i+1} \to \operatorname{coker} Fd^i \to 0$ remains exact, hence we have an isomorphism $F \operatorname{coker} d^i \cong \operatorname{coker} Fd^i$. Now, applying F to the exact sequence $0 \to \operatorname{im} d^i \to C^{i+1} \to \operatorname{coker} d^i \to 0$, we get an exact sequence

$$F \operatorname{im} d^i \to FC^{i+1} \to F \operatorname{coker} d^i \to 0$$

so by the universal property of kernels, we have an epimorphism

$$q : F \operatorname{im} d^i \to \ker F \operatorname{coker} d^i \cong \ker \operatorname{coker} Fd^i = \operatorname{im} Fd^i.$$

Finally, we have exact sequences

$$0 \to H^i(C) \to \operatorname{coker} d^{i-1} \to \operatorname{im} d^i \to 0$$

and

$$0 \to H^i(FC) \to \operatorname{coker} Fd^{i-1} \to \operatorname{im} Fd^i \to 0.$$

Applying F to the first, we obtain an exact sequence

$$FH^i(C) \to F \operatorname{coker} d^{i-1} \to F \operatorname{im} d^i \to 0.$$

We are now in the following situation:

$$\begin{tikzcd}
 & FH^i(C) \arrow[r] & F\operatorname{coker} d^{i-1} \arrow[r] \arrow[d, dashed, "\cong"] & F\operatorname{im} d^i \arrow[r] \arrow[d, "q"] & 0 \\
0 \arrow[r] & H^i(FC) \arrow[r] & \operatorname{coker} Fd^{i-1} \arrow[r] & \operatorname{im} Fd^i \arrow[r] & 0
\end{tikzcd}$$

where the blue composite is zero, hence there is a factoring $FH^i(C) \to H^i(FC)$ since the latter is the kernel of the final blue arrow. □

(b) If F is left exact, show that there is a natural morphism $FH^i(C) \leftarrow H^i(FC)$.

Proof. By the dual argument as above, or by 1.5.F, we have an isomorphism $F \ker d^i \cong \ker Fd^i$. Now apply F to the exact sequence $0 \to \ker d^i \to C^i \to \operatorname{im} d^i \to 0$ to obtain an exact sequence

$$0 \to F \ker d^i \to FC^i \to F \operatorname{im} d^i$$

hence $FC^i \to F \operatorname{im} d^i$ factors through the kernel to yield a monomorphism $j : \operatorname{im} Fd^i = \operatorname{coker}(F \ker d^i) \to F \operatorname{im} d^i$.

To finish, we have exact sequences

$$0 \to \operatorname{im} d^{i-1} \to \ker d^i \to H^i(C) \to 0$$

$$0 \to \operatorname{im} Fd^{i-1} \to \ker Fd^i \to H^i(FC) \to 0.$$

Applying F to the first, we get

$$0 \to F \operatorname{im} d^{i-1} \to F \ker d^i \to FH^i(C)$$

which yields the diagram

$$\begin{tikzcd}
0 \arrow[r] & \operatorname{im} Fd^{i-1} \arrow[r] \arrow[d, "j"] & \ker Fd^i \arrow[r] \arrow[d, "\cong"] & H^i(FC) \arrow[r] & 0 \\
0 \arrow[r] & F \operatorname{im} d^{i-1} \arrow[r] & F \ker d^i \arrow[r] & FH^i(C) &
\end{tikzcd}$$

where the red arrows compose to zero, implying that the composition of the last two arrows factors through $H^i(FC)$ by the universal property of the cokernel, thus yielding a map $H^i(FC) \to FH^i(C)$. ☐

(c) *If F is exact, show that the maps above are inverses, hence isomorphisms.*

Proof. If F is exact, then the two diagrams at the end of (a) and (b) become stacks of short exact sequences

$$\begin{tikzcd}
0 \arrow[r] & FH^i(C) \arrow[r] \arrow[d, dashed] & F \operatorname{coker} d^{i-1} \arrow[r] \arrow[d, "\cong"] & F \operatorname{im} d^i \arrow[r] \arrow[d, "q"] & 0 \\
0 \arrow[r] & H^i(FC) \arrow[r] & \operatorname{coker} Fd^{i-1} \arrow[r] & \operatorname{im} Fd^i \arrow[r] & 0
\end{tikzcd}$$

and

$$\begin{tikzcd}
0 \arrow[r] & \operatorname{im} Fd^{i-1} \arrow[r] \arrow[d, "j"] & \ker Fd^i \arrow[r] \arrow[d, "\cong"] & H^i(FC) \arrow[r] \arrow[d, dashed] & 0 \\
0 \arrow[r] & F \operatorname{im} d^{i-1} \arrow[r] & F \ker d^i \arrow[r] & FH^i(C) \arrow[r] & 0
\end{tikzcd}$$

implying they are isomorphisms. ☐

Exercise (1.5.J)**.** *Let $\mathcal{C}$ be an abelian category and $a, b : \mathcal{I} \to \mathcal{C}$ are diagrams. (Write $A_i = a(i), B_i = b(i)$). Let h be a natural transformation $a \to b$. Then the maps h_i yield another diagram in $\mathcal{C}$ indexed by $\mathcal{I}$. Assuming all limits exist, describe a canonical isomorphism* $\lim \ker(h_i) \to \ker(\lim A_i \to \lim B_i)$.

Proof. We have projections $\lim \ker(h_i)$ to each $\ker h_i$, which we can compose with the maps $\ker h_i \to A_i$ to yield maps $\lim \ker(h_i) \to A_i$ for each i whose composition with h_i is zero. Therefore we get a map $\lim \ker h_i \to \lim A_i$ whose composition with $\lim A_i \to \lim B_i$ is zero. Thus we get an induced map $\lim \ker h_i \to \ker(\lim A_i \to \lim B_i)$, say Φ.

We construct an inverse to Φ. First consider the monomorphism $\ker(\lim A_i \to \lim B_i) \to \lim A_i$. We can postcompose with the projection maps $\lim A_i \to A_i$ to obtain maps $\ker(\lim A_i \to \lim B_i) \to A_i$ which are zero after postcomposing with h_i. Hence each of these maps factors through $\ker h_i$. By the universal property of the limit these yields a map $\Psi : \ker(\lim A_i \to \lim B_i) \to \lim \ker h_i$.

Let us check that these are inverse. First for $\Psi \circ \Phi$. Let $\pi_i : \lim \ker h_i \to \ker h_i$ be the projections and $\iota_i : \ker h_i \to A_i$ be the inclusions. It suffices to show that $\pi_i \circ \Psi \circ \Phi = \pi_i$ for every i. The way we have defined Φ and Ψ means that they are the maps making the following diagram commute:

$$\begin{tikzcd}
 & \ker h_i \arrow[dl, bend right] & \lim \ker h_i \arrow[l, "\pi_i"'] \arrow[dl] \arrow[d, "\Phi"] \\
 & \lim A_i \arrow[dl] & \ker(\lim A_i \to \lim B_i) \arrow[l] \arrow[d, "\Psi"] \\
A_i & \ker h_i \arrow[l] & \lim \ker h_i \arrow[l, "\pi_i"]
\end{tikzcd}$$

which implies that $\iota_i \circ \pi_i \circ \Psi \circ \Phi = \iota_i \circ \pi_i$ for every i, whereupon we are done since ι_i is a monomorphism.

In the other direction, let j be the inclusion $\ker(\lim A_i \to \lim B_i) \to \lim A_i$. The diagram

$$\begin{tikzcd}
A_i & \lim A_i \arrow[l] & \ker(\lim A_i \to \lim B_i) \arrow[l] \arrow[dll, dashed] \arrow[d, dashed, "\Psi"] \\
\ker h_i \arrow[u] & & \lim \ker h_i \arrow[ll] \arrow[ul, dashed] \arrow[d, dashed, "\Phi"] \\
 & & \ker(\lim A_i \to \lim B_i) \arrow[uul]
\end{tikzcd}$$

tells us $j \circ \Phi \circ \Psi = j$, and since j is a mono we are done. □

Exercise (1.5.K)**.** *Make sense of the statement "limits commute with limits" in a general category and prove it.*

Proof. Let $\mathcal{I}$ and $\mathcal{J}$ be indexing categories, $\mathcal{C}$ an arbitrary category. Let $F : \mathcal{I} \times \mathcal{J} \to \mathcal{C}$ be a functor. Fixing $j \in \mathcal{J}$, we can consider the functors $i \mapsto F(i, j)$. This yields a functor $F(-,-)$ from $\mathcal{J}$ to $\mathsf{Fun}(\mathcal{I}, \mathcal{C})$, so we can consider the limit $\lim_{j\in\mathcal{J}} F(-, j)$ in this functor category. This limit is a functor $\mathcal{I} \to \mathcal{C}$, so we can again consider its limit $\lim_{i\in\mathcal{I}} F(-, j)(i) = \lim_{i\in\mathcal{I}} \lim_{j\in\mathcal{J}} F(i, j)$ which yields an object of $\mathcal{C}$. Likewise we could have started by fixing an element of $\mathcal{J}$ to get $\lim_{j\in\mathcal{J}} \lim_{i\in\mathcal{I}} F(i, j)$. The statement is that these two things are isomorphic. Let us prove this.

Let X be an element of $\mathcal{C}$ with maps $\pi_i : X \to \lim_j F(i, j)$ commuting with morphisms in $\mathcal{I}$. By the universal property of limits, each π_i is the data of maps $\pi_{i,j} : X \to F(i, j)$ commuting with morphisms in $\mathcal{J}$. But now these $\pi_{i,j}$ also give a family of maps $\pi_j : X \to \lim_i F(i, j)$ commuting with the morphisms in $\mathcal{J}$, upon which we see that X maps to $\lim_i F(i, j)$. So any cone over $i \mapsto \lim_j F(i, j)$ is a cone over $j \mapsto \lim_i F(i, j)$, so the terminal cones, i.e. the limits, are the same in both cases (if they exist). □

Exercise (1.5.L)**.** *Show that in* Mod_A*, colimits over filtered index categories are exact.*

Proof. We know that colimits commute with colimits, so colimits commute with cokernels. We just have to show that filtered colimits commute with kernels. Let $\mathcal{I}$ be a filtered indexing category and $F, G : \mathcal{I} \to \mathsf{Mod}_A$ be diagrams of A-modules. Let h be a natural transformation $F \to G$. Then we claim that $\operatorname{colim} \ker h_i = \ker(\operatorname{colim} F_i \to \operatorname{colim} G_i)$. Recall that the colimit of a diagram of modules is the direct sum over the diagram of the modules modulo the equivalence relation of "being in the image of one of the maps from the diagram." (Exercise 1.3.E, 1.3.F).

Let $x \in \ker(\operatorname{colim} F_i \to \operatorname{colim} G_i)$. Then x is represented by some $a \in F_i$ such that $h_i(a)$ is zero in $\operatorname{colim} G_i$. Let $0 \in G_k$. Then by definition of the colimit, there is some $j \in \mathcal{I}$ and morphisms $u : i \to j$ and $v : k \to j$ such that $Gu(h_i(a)) = Gv(0) = 0 \in G_j$. By naturality of h, this implies $h_j(Fu(a)) = 0$, so that $Fu(a) \in \ker h_j$. But $Fu(a)$ and a represent the same element in $\operatorname{colim} F_i$, hence $a \in \ker h_j$, so $x \in \operatorname{colim}(\ker h_i)$.

Conversely, if $y \in \operatorname{colim}(\ker h_i)$, then y is represented by some $b \in \ker h_i$ for some i, so $h_i(b) = 0$, so b represents an element in $\ker(\operatorname{colim} F_i \to \operatorname{colim} G_i)$. $\square$

Exercise (1.5.M)**.** *Show that filtered colimits commute with homology in* Mod_A.

Proof. Filtered colimits commute with colimits, so they are right exact. By the previous exercise they also commute with kernels, so they must commute with homology. $\square$

Exercise (1.5.N)**.** *Let* $(0 \to A_i \to B_i \to C_i \to 0)$ *be an inverse system of exact sequences of modules. Call the transition maps* $\alpha_i : A_i \to A_{i-1}$, $\beta_i : B_i \to B_{i-1}$, *and* $\gamma_i : C_i \to C_{i-1}$. *Assume that the* α_i *are surjective. Show that the limit* $0 \to \lim A_i \to \lim B_i \to \lim C_i \to 0$ *is also exact.*

Proof. To define the maps, we have $\lim A_i \to A_i \to B_i$ commuting with the structure maps, which induces a map $\lim A_i \to \lim B_i$, and likewise for $\lim B_i \to \lim C_i$. We are in the following situation:

$$\begin{tikzcd}
& \lim A_i \arrow[dl, "\pi^A_{i+1}"'] \arrow[dr] \arrow[dd, "f_*"] & \\
A_{i+1} \arrow[rr] \arrow[dd, "f_{i+1}"'] & & A_i \arrow[dd, "f_i"] \\
& \lim B_i \arrow[dl] \arrow[dr] \arrow[dd, "g_*"] & \\
B_{i+1} \arrow[rr] \arrow[dd, "g_{i+1}"'] & & B_i \arrow[dd, "g_i"] \\
& \lim C_i \arrow[dl] \arrow[dr] & \\
C_{i+1} \arrow[rr] & & C_i
\end{tikzcd}$$

Let us first show that $\lim A_i \to \lim B_i$ is injective. Let $x \in \lim A_i$ and suppose $f_*(x) = 0$. By commutativity of the diagram we have $f_i(\pi_i^A(x)) = 0$, so by injectivity of f_i we know $\pi_i^A(x) = 0$ for all i. It follows that $x = 0$.

For surjectivity of $\lim B_i \to \lim C_i$, take some $y \in \lim C_i$. Write c_i for the projections $\pi_i^C(y)$. By surjectivity of g_0, we can choose a lift b_0 of c_0. Inductively, given a lift b_i of c_i, choose a lift b'_{i+1} of c_{i+1}. Then

$$g_i(\beta_{i+1}(b'_{i+1}) - b_i) = \gamma_{i+1}(g_{i+1}(b'_{i+1})) - c_i = \gamma_{i+1}(c_{i+1}) - c_i = 0,$$

so $\beta_{i+1}(b'_{i+1}) - b_i \in \ker g_i = \operatorname{im} f_i$, so take some $a_i \in A_i$ mapping to it. By surjectivity of the α_i, there exists some a_{i+1} with $\alpha_{i+1}(a_{i+1}) = a_i$. Then the element $b_{i+1} := b'_{i+1} - f_{i+1}(a_{i+1})$ satisfies

$$g_{i+1}(b_{i+1}) = g_{i+1}(b'_{i+1}) - g_{i+1}(f_{i+1}(a_{i+1})) = c_i$$

and

$$\beta_{i+1}(b_{i+1}) = \beta_{i+1}(b'_{i+1}) - \beta_{i+1}(f_{i+1}(a_{i+1})) = \beta_{i+1}(b'_{i+1}) - f_i(\alpha_{i+1}(a_{i+1})) = b_i$$

so we have produced a $(b_i) \in \lim B_i$ mapping to y.

Finally, for exactness in the middle. Clearly $\operatorname{im} f_* \subset \ker g_*$, since any $x \in \lim A_i$ has to satisfy $\pi_i^C(g_* f_*(x)) = gf(\pi_i^A(x)) = 0$ for all i. Conversely, take some $z = \{b_i\} \in \ker g_*$. By commutativity, each $b_i \in \ker g_{i+1} = \operatorname{im} f_{i+1}$, so start with some $a_0 \in A_0$ with $f_0(a_0) = b_0$. Now inductively, given some a_i with $f_i(a_i) = b_i$, choose a lift a'_{i+1} of b_{i+1}. Then

$$f_i(\alpha_{i+1}(a'_{i+1}) - a_i) = \beta_i(f_{i+1}(a'_{i+1})) - b_i = \beta_i(b_{i+1}) - b_i = 0,$$

so $\alpha_{i+1}(a'_{i+1}) - a_i$ is in $\ker f_i = 0$, so actually $\alpha_{i+1}(a'_{i+1}) = a_i$, so we have found an element $\{a_i\} \in \lim A_i$ mapping to z, showing that $\operatorname{im} f_* = \ker g_*$. $\square$

1.6. **Spectral sequences (tbd).**

> "You should not read this section when you are reading the rest of Chapter 1. Instead, you should read it just before you need it for the first time."

2. Sheaves

2.1. Motivating example: the sheaf of smooth functions.

Exercise (2.1.A). *Show that $\mathfrak{m}_p$ is the only maximal ideal of $\mathcal{O}_p$.*

Proof. It suffices to show that any f not in $\mathfrak{m}_p$ is invertible. If $f \notin \mathfrak{m}_p$, then f is nonzero at p, so by smoothness there is some open subset U around p where f is also nonzero, hence $1/f$ is a well-defined function on U. Then $(1/f, U)$ is an inverse to f in $\mathcal{O}_p$. □

Exercise (2.1.B). *Let X be a differentiable manifold, $p \in X$, $\mathcal{O}_p$ and $\mathfrak{m}_p$ as above. Prove that $\mathfrak{m}_p/\mathfrak{m}_p^2$ is naturally the cotangent space to X at p.*

Proof. One way to define the tangent space of X at p is as the space of derivations at p:

$$T_pX = \{D : C^\infty(X) \to \mathbf{R} : D(fg) = f(p)D(g) + g(p)D(f)\}$$

so that the cotangent space T_p^*X is just the dual $\mathrm{Hom}_{\mathbf{R}}(T_pX, \mathbf{R})$. Consider the map $\phi : \mathfrak{m}_p \to T_p^*X$ defined as follows. Given a germ $(f, U) \in \mathfrak{m}_p$, we can extend $f : U \to \mathbf{R}$ to a function $\tilde{f} : X \to \mathbf{R}$ via bump functions. Then $(D \mapsto D(\tilde{f}))$ yields an element in T^*pX, which we define to be $\phi(f, U)$. This is well-defined for the following reason. If (f, U) and (g, V) represent the same germ, then there is some neighborhood of p in which $f - g$ is zero. We can define a bump function b to be one inside this neighborhood where $f - g$ is zero, and zero outside. Then $b(f - g)$ is identically zero on X. Then $0 = D(b(f - g)) = b(p)D(f - g) + D(b)(f - g)(p) = D(f - g)$.

Clearly if $f, g \in \mathfrak{m}_p$, then $\phi(fg) = D(\tilde{f}\tilde{g}) = f(p)D(\tilde{g}) + g(p)D(\tilde{f}) = 0$, so ϕ descends to a map $\mathfrak{m}_p/\mathfrak{m}_p^2 \to T_p^*X$. ... tbd □

2.2. Definition of sheaf and presheaf.

Exercise (2.2.A). *Given a topological space X, let $\mathsf{Open}(X)$ be the category of open sets where the morhpisms are inclusions. Show that the data of a presheaf is the data of a contravariant functor from $\mathsf{Open}(X)$ to the category of sets.*

Proof. Let us start with a presheaf $\mathcal{F}$. We can define a functor $F : \mathsf{Open}(X) \to \mathsf{Set}$ by sending an open set U to $\mathcal{F}(U)$, and an inclusion $U \subset V$ to the restriction map $\mathrm{res}_{V,U} : \mathcal{F}(V) \to \mathcal{F}(U)$. Then the functor respects composition since the restriction maps do, and the identity inclusion maps to the identity map. Conversely, given a functor $F : \mathsf{Open}(X)^{\mathrm{op}} \to \mathsf{Set}$, we can define a presheaf $\mathcal{F}$ via $\mathcal{F}(U) = F(U)$. For an inclusion $i : U \to V$ we define $\mathrm{res}_{V,U}$ to be $F(i)$. Then the definition of a functor (identity, composition) give us the presheaf conditions. □

Exercise (2.2.B). *Show that the following are presheaves on $\mathbf{C}$ with the standard topology, but not sheaves:*

(a) *Bounded functions.*

Proof. Take an open cover of $\mathbf{C}$ by open balls $U_n = \{|z| < n\}$. Then the identity map is an element of each $\mathcal{F}(U_n)$. But there is no bounded function that on $\mathbf{C}$ that these glue to. □

(b) *Holomorphic functions admitting a holomorphic square root.*

Proof. Recall the argument principle: if f is a meromorphic function inside and on a closed contour C, then $\int_C f'/f$ is some integer multiple (number of zeros minus number of poles) of $2\pi i$. Take C to be the unit circle and let f be the identity function. We claim that f has no square root in the annulus $A = \{1 - \epsilon < |z| < 1 + \epsilon\}$ for some $0 < \epsilon < 1$. Suppose otherwise, i.e.

there is some holomorphic g defined on A with $g^2 = f$. Then by the argument principle, we have

$$2\pi i = \int_C \frac{f'(z)}{f(z)} dz = 2 \int_C \frac{g'(z)}{g(z)} dz \in 4\pi i \mathbf{Z}$$

which is a contradiction. Thus $f(z) = z$ admits no holomorphic square root in A.But now we can cover the annulus A with simply connected regions, on each of which we can define a holomorphic square root of f. So on each of these regions $f(z) = z$ restricts to the identity, so they agree on all intersections, but cannot patch together to a function which admits a holomorphic square root. □

Exercise (2.2.C)**.** *The identity and gluability axioms for a sheaf may be interpreted as saying that $\mathcal{F}(\cup_i U_i)$ is a certain limit. What limit?*

Solution. Let $\mathcal{F}$ be a sheaf over X and let U_i be an open cover of open $U \subset X$. This answer has to do with the "equalizer exact sequence" in the paragraph preceeding the exercise.

Consider the diagram in Set whose objects are the $\mathcal{F}(U_i)$ as well as pairwise intersections $\mathcal{F}(U_i \cap U_j)$ for all i, j. We claim that $\mathcal{F}(U) = \mathcal{F}(\cup_i U_i)$ is the limit of this diagram, where the structure maps are induced by the inclusions in $\mathsf{Open}(X)$, which obviously commute. Let G be another cone of this diagram with structure maps $\phi_i : G \to \mathcal{F}(U_i)$ and $\phi_{ij} : G \to F(U_i \cap U_j)$. Then for any $g \in G$, the images $\phi_i(g)$ are elements of each $\mathcal{F}(U_i)$ whose restrictions to each intersection agree, which by the glueabilty and identity axioms yields a global element in $\mathcal{F}(U)$, which yields our function $\phi : G \to \mathcal{F}(U)$. The axioms further tell us that the restriction of $\phi(g)$ to each U_i agree with the $\phi_i(g)$. This shows that $\mathcal{F}(U)$ is the limit. □

Exercise (2.2.D)**.** *Sheaf checking.*

(a) Verify that the following are sheaves: smooth functions, continuous functions, real-analytic, real-valued functions on a manifold X.

Solution. It is easy to see that these are presheaves: the restriction of any of these classes of functions to a smaller open set preserves their smoothness/continuity/analyticity/real-valuedness, and the rules of function composition and restriction yield functoriality.

To see these satisfy the identity axiom, let U be an open set in X, $\{U_i\}$ an open cover of it. Let $\mathcal{F}$ be the presheaf of any of the above over X. Suppose $f, g \in \mathcal{F}(U)$ restrict to the same function on all U_i. Then for any $x \in U$, it belongs to some U_i, whereby $f(x) = f|_{U_i}(x) = g|_{U_i}(x) = g(x)$.

For the glueability axiom, again let $\{U_i\}$ be an open cover of U and suppose we are given $f_i \in \mathcal{F}(U_i)$ that agree on the intersections. We wish to define a function f on U that restricts to each of the f_i. So for each $x \in U$, choose some U_i which x belongs to and define $f(x) = f_i(x)$. This is well-defined since the f_i agree on intersections. And f remains smooth/continuous/analytic/real-valued since these are local properties. □

(b) Show that real-valued continous functions on a topological space X form a sheaf.

Proof. Call this $\mathcal{F}$. Again, $\mathcal{F}$ is a presheaf by virtue of function composition and restriction: if $U \subset V$ and $f : V \to X$ is continuous, then so is $f|_U : U \to X$, since the preimage of an open set $A \subset X$ is just $f|_U^{-1}(A) = U \cap f^{-1}(A)$, which is open in the subspace topology. The same argument as above gives the identity axiom, and the glueability axiom comes from the pasting lemma (which says that the f constructed in (a) is continuous). □

Exercise (2.2.E)**.** *Let X be a topological space and let S be a set. Let $\mathcal{F}(U)$ be the set of maps $U \to S$ that are locally constant, i.e. for any $p \in U$ there is an open neighborhood around p where the map is constant. Show that $\mathcal{F}$ is a sheaf on X.*

Proof. The restriction of a locally constant function is clearly locally constant, and the restriction of a restriction is just restriction, so $\mathcal{F}$ is a presheaf.

The identity axiom is the same as above. For the glueability axiom, given $f_i \in \mathcal{F}(U_i)$ and define the glued f the same as above. Then for any $x \in U$, there is some U_i containing x and a neighborhood V of x contained in U_i where $f_i|_V = f|_V$ is constant. □

Exercise (2.2.F)**.** *Suppose Y is a topological space. Show that continuous maps to Y form a sheaf of sets on a space X. (This is a generalization of the previous exercises).*

Proof. There is nothing new here, the proof is exactly the same. □

Exercise (2.2.G)**.** *Fancier sheaves.*

(a) Show that the sections of a continuous map $\mu : Y \to X$ yield a sheaf on X.

Proof. For $U \subset X$ open we let $\mathcal{F}(U)$ be those maps $s : U \to Y$ for which $s \circ \mu = \mathrm{id}|_U$. Let us first show this is a presheaf on $\mathsf{Open}(X)$. To an inclusion $U \subset V$ we define the restriction map $\mathrm{res}_{V,U}(f) = f|_U$, which remains a section since $(f|_U \circ \mu)(x) = (f \circ \mu)(x) = x$. And restrictions of restrictions are restrictions, so we are done.

For the identity axiom, again let U be open in X and $\{U_i\}$ an open cover of it. Suppose $f|_{U_i} = g|_{U_i}$ for all i. Then for any $x \in U$, it belongs to some U_i, upon which $f = g$ on all of U. For glueability, let $\{U_i\}$ be same as above and suppose we have sections $f_i \in \mathcal{F}(U_i)$ agreeing on intersections. Then define $f : U \to Y$ via the pasting lemma. Then $(f \circ \mu)(x) = (f_i \circ \mu)(x)$ for some i, showing that f is a section. □

(b) Suppose $(Y, +)$ is a topological group. Show that continuous maps $X \to Y$ form a sheaf of groups.

Proof. The group structure on $\mathcal{F}(U)$ is given by letting $f + g$ be the map $x \mapsto f(x) + g(x)$. The identity is the map sending all elements of X to $0 \in Y$, and the inverse of a map f is the map $x \mapsto -f(x)$. For an inclusion $U \subset V$ again define $\mathrm{res}_{V,U}(f) = f|_U$, which remains continuous. This is a group homomorphism since $(f + g)|_U = f|_U + g|_U$. Restrictions of restrictions are restrictions, so this makes $\mathcal{F}$ into a presheaf of groups.

The identity axiom is the same as above. For glueability, let $\{U_i\}$ be an open cover of open $U \subset X$ and choose f_i in each $\mathcal{F}(U_i)$ agreeing on intersections. Define $f : U \to Y$ again via the pasting lemma. □

Exercise (2.2.H)**.** *Let $\pi : X \to Y$ be a continuous map, and let $\mathcal{F}$ be a presheaf on X. Define a presheaf $\pi_*\mathcal{F}$ by $\pi_*\mathcal{F}(V) = \mathcal{F}(\pi^{-1}(V))$. Show this is a presheaf, and a sheaf if $\mathcal{F}$ is a sheaf.*

Proof. Let us first show it is a presheaf. Given an inclusion $W \subset V$ of open sets of Y, we know $\pi^{-1}(W) \subset \pi^{-1}(V)$ are open in X, so we can define $\mathrm{res}_{W,V}(f) = \mathrm{res}_{\pi^{-1}(W),\pi^{-1}(V)}(f)$, upon which composition and identity follow from the fact that $\mathcal{F}$ is a presheaf.

Now suppose $\mathcal{F}$ is a sheaf. Suppose we have an open cover $\{V_i\}$ of V. Then $\{\pi^{-1}(V_i)\}$ is an open cover of $\pi^{-1}(V)$. So if $f, g \in \pi_*\mathcal{F}(V) = \mathcal{F}(\pi^{-1}(V))$ agree on all their restrictions to $\pi^{-1}(V_i)$, then by the identity axiom on $\mathcal{F}$, we know $f = g$ in $\pi_*\mathcal{F}(V) = \mathcal{F}(\pi^{-1}(V))$. Likewise, for glueability, if we are given compatible $f_i \in \pi_*\mathcal{F}(V_i) = \mathcal{F}(\pi^{-1}(V_i))$, then they glue to an element in $\pi_*\mathcal{F}(V) = \mathcal{F}(\pi^{-1}(V))$ as desired. □

Exercise (2.2.I)**.** *Let $\pi : X \to Y$ be a continuous map, and $\mathcal{F}$ a sheaf of sets (or rings or modules) on X. if $\pi(p) = q$, describe the natural morphism of stalks $(\pi_*\mathcal{F})_q \to \mathcal{F}_p$.*

Solution. The first approach is to use the explicit definition of stalks via representatives, the second via colimits.

For the first, recall that an element of $(\pi_*\mathcal{F})_q$ is represented by a tuple (f, V) where $f \in \pi_*\mathcal{F}(V) = \mathcal{F}(\pi^{-1}(V))$ and $q \in V$. It follows that $p \in \pi^{-1}(V)$, so we can send (f, V) to $(f, \pi^{-1}(V))$. Let us see this is well-defined. Suppose $(f, V) \sim (g, W)$, i.e. there is some open $U \subset V, W$ such that $\mathrm{res}_{U,V}(f) = \mathrm{res}_{U,W}(g)$. By how we have defined the pushforward sheaf, $\mathrm{res}_{\pi^{-1}(U),\pi^{-1}(V)}(f) = \mathrm{res}_{\pi^{-1}(U),\pi^{-1}(W)}(g)$, so that $(f, \pi^{-1}(V)) \sim (g, \pi^{-1}(W))$.

If we work with colimits, notice that for any open $V \subset Y$ containing q, that elements of $\pi_*\mathcal{F}(V) = \mathcal{F}(\pi^{-1}(V))$ are representatives in the colimit $\mathrm{colim}_{U \ni p}(\mathcal{F}(U)) = \mathcal{F}_p$, so we have maps from each $\pi_*(\mathcal{F}(V))$ to $\mathcal{F}_p$, which are compatible with the restriction maps (since the restriction maps for $\pi_*\mathcal{F}$ are just those of $\mathcal{F}$). Thus by the universal property of colimits this yields a unique map $(\pi_*\mathcal{F})_q = \mathrm{colim}_{V \ni q}(\pi_*\mathcal{F}(V)) \to \mathrm{colim}_{U \ni p}(\mathcal{F}(U)) = \mathcal{F}_p$. □

Exercise (2.2.J)**.** *If $(X, \mathcal{O}_X)$ is a ringed space, and $\mathcal{F}$ is an $\mathcal{O}_X$-module, describe how for each $p \in X$, $\mathcal{F}_p$ is an $\mathcal{O}_{X,p}$-module.*

Solution. Take $(g, V) \in \mathcal{O}_{X,p}$ and $(f, U) \in \mathcal{F}_p$. Then $W := V \cap U$ is open, so we can consider the restrictions $g|_W \in \mathcal{O}_X(W)$ and $f|_W \in \mathcal{F}(W)$, whereupon the $\mathcal{O}_X$-module structure gives an element $(g|_W) \cdot (f|_W) \in \mathcal{F}(W)$. Then define $(g, V) \cdot (f, U)$ to be $((g|_W) \cdot (f|_W), W) \in \mathcal{F}_p$.

To see this is well defined, take other representatives (g', V') and (f', U'), with $g'|_{V''} = g|_{V''}$ and $f'|_{U''} = f|_U$. Let $W' = U' \cap W'$ and $W'' = U'' \cap W''$. Then the restrictions of $(g|_W) \cdot (f|_W)$ and $(g'|_{W'}) \cdot (f'|_{W'})$ to W'' are the same by the $\mathcal{O}_X$-module structure. The module axioms follow from the fact that $\mathcal{F}$ is an $\mathcal{O}_X$-module and working with intersections of open sets. □

2.3. **Morphisms of presheaves and sheaves.**

Exercise (2.3.A)**.** *If $\phi : \mathcal{F} \to \mathcal{G}$ is a morphism of presheaves on X, describe an induced morphism of stalks $\phi_p : \mathcal{F}_p \to \mathcal{G}_p$. (So that taking the stalk at a point p yields a functor from the category of presheaves over X to the category of sets.)*

Solution. We define $\phi_p(f, U) := (\phi(U)(f), U)$. This is well-defined, for if we have another representative (g, V) such that $f|_W = g|_W$ for $W \subset U, V$, then by naturality of ϕ, we get that $\phi(U)(f)|_W = \phi(U)(f|_W) = \phi(U)(g|_W) = \phi(U)(g)|_W$.

On the other hand, using the colimit definition of a stalk, for each $U \in \mathsf{Open}(X)$, we have a map $\phi(U) : \mathcal{F}(U) \to \mathcal{G}(U) \to \mathrm{colim}_{p \in U} \mathcal{G}U$ which commutes with the restriction maps by definition of ϕ, so by the universal property of colimits this induces the desired map $\mathcal{F}_p \to \mathcal{G}_p$. □

Exercise (2.3.B)**.** *Suppose $\pi : Y \to X$ is a continuous map of topological spaces. Show that pushforward gives a functor $\mathsf{Set}_X \to \mathsf{Set}_Y$.*

Proof. We have already defined this on the objects, so we need to specify what happens to morphisms. Let $\phi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves on X. Then we need to define a family of maps $\pi_*\phi : \pi_*\mathcal{F}(V) \to \pi_*\mathcal{G}(V)$ for each open $V \subset Y$. But this is the same thing as defining maps $\mathcal{F}(\pi^{-1}(V)) \to \mathcal{G}(\pi^{-1}(V))$, and we can just define these to be $\pi_*\phi(V) := \phi(\pi^{-1}(V))$. Then by the fact that ϕ is a morphism of presheaves, so is $\pi_*\phi$. For functoriality, clearly if ϕ is the identity then so is $\pi_*\phi$, and given another morphism $\psi : \mathcal{G} \to \mathcal{H}$ of presheaves over X, we get that $(\pi_*\psi \circ \pi_*\phi)(V) = (\psi \circ \pi)(\pi^{-1}(V)) = \psi(\pi^{-1}(V)) \circ \phi(\pi^{-1}(V)) = \pi_*\psi(V) \circ \psi_*\phi(V)$. □

Exercise (2.3.C)**.** *Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves of sets on X. Show that "sheaf hom," $\mathcal{H}om(\mathcal{F}, G)(U) := \mathrm{Mor}(\mathcal{F}|_U, \mathcal{G}|_U)$, is a sheaf on X.*

Proof. Let $U \subset V$ be an inclusion of open sets in X. We need to define the restriction function $\mathrm{Mor}(\mathcal{F}|_V, \mathcal{G}|_V) \to \mathrm{Mor}(\mathcal{F}|_U, \mathcal{G}|_U)$. Given a natural transformation $\alpha \in \mathrm{Mor}(\mathcal{F}|_V, \mathcal{G}|_V)$, we have for any open set W in U (and hence in V) a map $\alpha(W) : \mathcal{F}(W) \to \mathcal{G}(W)$. So just send α to $\alpha|_U$. Then the identity gets sent to the identity, and composition is respected by restriction.

For the identity axiom. Suppose $\{U_i\}$ is an open cover of U and that $f, g \in \mathcal{H}om(\mathcal{F}, \mathcal{G})(U)$ agree on all their restrictions $f_i := f|_{U_i}, g_i = g|_{U_i} \in \mathrm{Mor}(\mathcal{F}|_{U_i}, \mathcal{G}|_{U_i})$, that is, for any open set $V_i \subset U_i$, $f_i(V_i)$ and $g_i(V_i)$ are the same function. Now take an element $x \in \mathcal{F}(U)$. We then get the commutative diagram

$$\begin{tikzcd} \mathcal{F}(U) \arrow[r, shift left, "f(U)"] \arrow[r, shift right, "g(U)"'] \arrow[d, "\mathrm{res}^{\mathcal{F}}_{U,U_i}"'] & \mathcal{G}(U) \arrow[d, "\mathrm{res}^{\mathcal{G}}_{U,U_i}"] \\ \mathcal{F}(U_i) \arrow[r, "f(U)|_{U_i} = g(U)|_{U_i}"'] & \mathcal{G}(U_i) \end{tikzcd}$$

which implies $\mathrm{res}^{\mathcal{G}}_{U,U_i}(f(U)(x)) = \mathrm{res}^{\mathcal{G}}_{U,U_i}(g(U)(x))$ for all i, and because $\mathcal{G}$ is a sheaf this implies $f(U)(x) = g(U)(x)$.

For glueability. Let $\{U_i\}$ be an open cover of U and suppose we have natural transformations $\alpha_i \in \mathcal{H}om(\mathcal{F}, \mathcal{G})(U_i)$ for each i that agree on intersections. We would like to define an element $\alpha \in \mathcal{H}om(\mathcal{F}, \mathcal{G})(U) = \mathrm{Mor}(\mathcal{F}|_U, \mathcal{G}|_U)$ (i.e. a natural family of functions $\mathcal{F}(V) \to \mathcal{G}(V)$ for each open $V \subset U$), that restricts to each U_i. So fix an open subset $V \subset U$ and an element $x \in \mathcal{F}(V)$. Then $V_i = V \cap U_i$ yields an open cover of V. Consider the elements

$$\alpha_i(V_i)(\mathrm{res}^{\mathcal{F}}_{V,V_i}(x)) \in \mathcal{G}(V_i)$$

which are a family of elements in each $\mathcal{G}(V_i)$ that agree on the intersections of the V_i, so because $\mathcal{G}$ is a sheaf, this yields an element in $\mathcal{G}(V)$ which we shall define to be $\alpha(V)(x)$. Then the restriction of α to U_i is α_i by construction. □

Exercise (2.3.D)**.** *More sheaf homs.*

(a) If $\mathcal{F}$ is a sheaf of sets on X, show that $\mathcal{H}om(\underline{\{*\}}, \mathcal{F}) \cong \mathcal{F}$, where $\underline{\{*\}}$ is the constant sheaf with values in the one-element set $\{*\}$.

Proof. For any open set U, we have $\mathcal{H}om(\underline{\{*\}}, \mathcal{F})(U) = \mathrm{Mor}(\{*\}, \mathcal{F}(U))$, which is in natural bijection $\mathcal{F}(U)$ just by looking at where $*$ goes. □

(b) If $\mathcal{F}$ is a sheaf of abelian groups on X, show that $\mathcal{H}om_{\mathsf{Ab}_X}(\underline{\mathbf{Z}}, \mathcal{F}) \cong \mathcal{F}$ as an isomorphism of sheaves of abelian groups.

Proof. Again, for open U, we have $\mathcal{H}om_{\mathsf{Ab}_X}(\underline{\mathbf{Z}}, \mathcal{F})(U) = \mathrm{Hom}(\mathbf{Z}, \mathcal{F}(U))$, which is naturally isomorphic to $\mathcal{F}(U)$ by looking at where $1 \in \mathbf{Z}$ is sent. □

(c) If $\mathcal{F}$ is an $\mathcal{O}_X$-module, show that $\mathcal{H}om_{\mathsf{Mod}_{\mathcal{O}_X}}(\mathcal{O}_X, \mathcal{F}) \cong \mathcal{F}$ as $\mathcal{O}_X$-modules.

Proof. Same as above, we have $\mathcal{H}om_{\mathsf{Mod}_{\mathcal{O}_X}}(\mathcal{O}_X, \mathcal{F})(U) = \mathrm{Hom}(\mathcal{O}_X(U), \mathcal{F}(U)) \cong \mathcal{F}(U)$ naturally via the map $f \mapsto f(1)$. □

Exercise (2.3.E)**.** *If $\phi : \mathcal{F} \to \mathcal{G}$ is a morphism of presheaves (of abelian groups) on X, show that* $\ker \phi$ *given by* $(\ker \phi)(U) = \ker(\phi(U))$ *is a presheaf.*

Proof. Let $V \subset U$ be open subsets of X. Then we have exact sequences assembling into the following commutative diagram:

$$\begin{tikzcd} 0 \arrow[r] & \ker(\phi(U)) \arrow[r] \arrow[d, dashed] & \mathcal{F}(U) \arrow[r, "\phi(U)"] \arrow[d, "\mathrm{res}^{\mathcal{F}}_{V,U}"] & \mathcal{G}(U) \arrow[d, "\mathrm{res}^{\mathcal{G}}_{V,U}"] \\ 0 \arrow[r] & \ker(\phi(V)) \arrow[r] & \mathcal{F}(V) \arrow[r, "\phi(V)"] & \mathcal{G}(V) \end{tikzcd}$$

By exactness and commutativity of the diagram, the composition of the two red arrows, after post-composing with $\phi(V)$, yields the zero map, so by the universal property it has to factor through $\ker(\phi(V))$ via the dotted map above. Let this dotted arrow be $\mathrm{res}^{\ker\phi}_{V,U}$.

Then given another open subset $W \subset V$, we can just stack another row below the diagram above, and it follows by uniqueness that $\mathrm{res}^{\ker\phi}_{W,V} \circ \mathrm{res}^{\ker\phi}_{V,U}$ is equal to $\mathrm{res}^{\ker\phi}_{W,U}$ since the former is a factoring of $\mathrm{res}^{\mathcal{F}}_{W,V} \circ \mathrm{res}^{\mathcal{F}}_{V,U}$ and the latter is a factoring of $\mathrm{res}^{\mathcal{F}}_{W,U}$, but these two are the same since $\mathcal{F}$ is a presheaf. And clearly $\mathrm{res}^{\ker\phi}_{U,U}$ is the identity. □

Exercise (2.3.F)**.** *Let $\phi : \mathcal{F} \to \mathcal{G}$ be as above. Show that* $\operatorname{coker}\phi$ *satisfies the universal property of cokernels in the category of presheaves.*

Proof. Let $\psi : \mathcal{G} \to \mathcal{H}$ be a morphism of presheaves such that $\psi \circ \phi = 0$, i.e. $\psi(U) \circ \phi(U) = 0$ for all opens $U \subset X$. Let $q(U) : \mathcal{G}(U) \to (\operatorname{coker}\phi)(U)$ be the canonical cokernel maps. By the universal property of cokernels in the category of abelian groups, there is a unique map $j(U) : \operatorname{coker}(\phi(U)) = (\operatorname{coker}\phi)(U) \to \mathcal{H}(U)$ such that $\psi(U) = j(U) \circ q(U)$. It remains to show that the collection of $j(U)$ for each U assemble into a morphism of presheaves $\operatorname{coker}\phi \to \mathcal{H}$. So let $V \subset U$ be an inclusion. We then get the diagram below:

$$\begin{tikzcd}
 & & \operatorname{coker}(\phi(U)) \arrow[d, dashed, "j(U)"] \arrow[dr, dashed, "\mathrm{res}^{\mathrm{coker}}"] & \\
\mathcal{F}(U) \arrow[r] \arrow[dr, "\mathrm{res}^{\mathcal{F}}"'] & \mathcal{G}(U) \arrow[r] \arrow[ur, "q(U)"] \arrow[dr, "\mathrm{res}^{\mathcal{G}}"'] & \mathcal{H}(U) \arrow[dr, "\mathrm{res}^{\mathcal{H}}"] & \operatorname{coker}(\phi(V)) \arrow[d, dashed, "j(V)"] \\
 & \mathcal{F}(V) \arrow[r] & \mathcal{G}(V) \arrow[r] \arrow[ur, "q(V)"'] & \mathcal{H}(V)
\end{tikzcd}$$

On the right "triangular prism" looking thing, the bottom square (consisting of $\mathcal{G}$, $\mathcal{H}$) commutes, and the diagonal square (consisting of $\mathcal{G}$, $\operatorname{coker}\phi$) commutes. We then get the following sequence of equalities:

$$\begin{aligned}
\mathrm{res}^{\mathcal{H}} \circ j(U) \circ q(U) &= \mathrm{res}^{\mathcal{H}} \circ \psi(U) \\
&= \psi(V) \circ \mathrm{res}^{\mathcal{G}} \\
&= j(V) \circ q(V) \circ \mathrm{res}^{\mathcal{G}} \\
&= j(V) \circ \mathrm{res}^{\mathrm{coker}} \circ q(U)
\end{aligned}$$

and because $q(U)$ is an epimorphism we conclude the the back square commutes, i.e. j is a morphism of presheaves which satisfies $\psi(U) = j(U) \circ q(U)$ for all U. Uniqueness follows from uniqueness in the category of abelian groups for each U. □

Exercise (2.3.G)**.** *Show that for a topological space X and open $U \subset X$, $\mathcal{F} \mapsto \mathcal{F}(U)$ is a functor from* $\mathsf{Ab}^{\mathrm{pre}}_X$ *to* Ab*. Then show that this functor is exact.*

Proof. For a morphism of sheaves $\phi : \mathcal{F} \to \mathcal{G}$ we automatically get a map $\phi(U) : \mathcal{F}(U) \to \mathcal{G}(U)$. For another morphism $\psi : \mathcal{G} \to \mathcal{H}$, we get $(\psi \circ \phi)(U) = \psi(U) \circ \phi(U)$ by definition, and the identity presheaf morphism $\mathrm{id} : \mathcal{F} \to \mathcal{F}$ is also the identity on each open set. So this is a functor.

By the way we have defined the kernel and cokernel of a presheaf, we know that $(\ker\phi)(U) = \ker(\phi(U))$ and likewise for the cokernel. So this functor preserves (co)kernels and is therefore exact. □

Exercise (2.3.H)**.** *Show that a sequence of presheaves* $0 \to \mathcal{F}_1 \xrightarrow{\phi_1} \cdots \xrightarrow{\phi_{n-1}} \mathcal{F}_n \to 0$ *is exact iff* $0 \to \mathcal{F}_1(U) \to \cdots \to \mathcal{F}_n(U) \to 0$ *is exact for all* U.

Proof. The forward direction follows from the previous exercise. Conversely, we have that

$$(\ker \phi_i)(U) = \ker(\phi_i(U)) = \operatorname{im}(\phi_{i-1}(U)) = (\operatorname{im} \phi_{i-1})(U)$$

for all U and i, so $\ker \phi_i = \operatorname{im} \phi_{i-1}$ for all i. (Here ϕ_0 and ϕ_n are the zero maps.) □

Exercise (2.3.I)**.** *Let* $\phi : \mathcal{F} \to \mathcal{G}$ *be a morphism of sheaves on* X. *Show that the presheaf kernel is also a sheaf, and that it satifes the universal property.*

Proof. We already know it is a presheaf. For the identity axiom, let U be an open set and $\{U_i\}$ an open cover. Suppose we have two elements $f, g \in \ker_{\mathrm{pre}} \phi$ whose restrictions $f|_{U_i}$ and $g|_{U_i}$ agree for all i. Let $\iota : \ker_{\mathrm{pre}} \phi \to \mathcal{F}$ denote the canonical monomorphism. By this being a morphism of presheaves, we have

$$\iota(U)(f)|_{U_i} = \iota(U_i)(f|_{U_i}) = \iota(U_i)(g|_{U_i}) = \iota(U)(g)|_{U_i}$$

for all i, and because $\mathcal{G}$ is a sheaf we get by the identity axiom that $\iota(U)(f) = \iota(U)(g)$. But $\iota(U)$ is injective, so $f = g$.

For glueability, again let U be open and $\{U_i\}$ an open cover. Suppose we have $f_i \in \ker_{\mathrm{pre}} \phi(U_i)$ agreeing on all intersections. Then

$$\iota(U_i)(f_i)|_{U_i \cap U_j} = \iota(U_i \cap U_j)(f_i|_{U_i \cap U_j}) = \iota(U_i \cap U_j)(f_j|_{U_i \cap U_j}) = \iota(U_j)(f_j)|_{U_i \cap U_j}$$

so the $\iota(U_i)(f_i)$ are also a compatible family of sections in $\mathcal{F}$, so by glueability there is some $\tilde{f} \in \mathcal{F}(U)$ such that $\tilde{f}|_{U_i} = \iota(U_i)(f_i)$ for all i. Notice that for each U_i,

$$\phi(U)(\tilde{f})|_{U_i} = \phi(U_i)(\tilde{f}|_{U_i}) = \phi(U_i)(f_i) = 0$$

since $f_i \in \ker_{\mathrm{pre}} \phi(U_i)$. Therefore $\phi(U)(\tilde{f})|_{U_i}$ is zero for all i, so $\phi(U)(\tilde{f}) = 0$ and there exists some $f \in \ker_{\mathrm{pre}} \phi(U)$ with $\iota(U)(f) = \tilde{f}$, from which it follows that

$$\iota(U_i)(f|_{U_i}) = \iota(U)(f)|_{U_i} = \tilde{f}|_{U_i} = \iota(U_i)(f_i)$$

and because $\iota(U_i)$ is a mono we get $f|_{U_i} = f_i$ for all i.

Suppose $\psi : \mathcal{E} \to \mathcal{F}$ is a morphism of sheaves with $\psi \circ \phi = 0$. Then $\psi(U)$ factors through $\ker_{\mathrm{pre}} \phi(U) = \ker(\phi(U))$ for each U, yielding a morphism of presheaves by 2.3.E. But a morphism of sheaves is just a morphism of underlying presheaves. This yields the universal property. □

Exercise (2.3.J)**.** *Let* X *be* $\mathbf{C}$ *with the standard topology,* $\mathcal{O}_X$ *be the sheaf of holomorphic functions, and let* $\mathcal{F}$ *be the presheaf of functions admitting a holomorphic logarithm. Describe an exact sequence of presheaves on* X

$$0 \to \underline{\mathbf{Z}} \to \mathcal{O}_X \to \mathcal{F} \to 0$$

but show that $\mathcal{F}$ *is not a sheaf. (Hence the presheaf cokernel is not always the sheaf cokernel.)*

Proof. The map $\phi : \underline{\mathbf{Z}} \to \mathcal{O}_X$ is given by inclusion: each map $\mathbf{Z} = \underline{\mathbf{Z}}(U) \to \mathcal{O}_X(U)$ sends $n \in \mathbf{Z}$ to the constant function at n. This is clearly injective. The map $\psi : \mathcal{O}_X \to \mathcal{F}$ sends $f \in \mathcal{O}_X(U)$ to $\exp(2\pi i f) \in \mathcal{F}(U)$. This is clearly surjective, since if h admits a holomorphic logarithm then $\psi(\frac{\log h}{2\pi i}) = \exp(\log h) = h$. For exactness, we clearly have in one direction $\psi(U) \circ \phi(U)(n) = \exp(2\pi i n) = 1$. In the other, suppose $\exp(2\pi i f) = 1$. Then $f(z)$ is an integer for all $z \in U$, so f, being holomorphic, must be locally constant on U, i.e. $f \in \underline{\mathbf{Z}}(U)$.

But $\mathcal{F}$ is not a sheaf. Cover $\mathbf{C}^\times$ with two open sets U_1, U_2 and choose a branches of $\log z$ on each. Then $h(z) = z$ restricted to each U_i admits a holomorphic logarithm, but the only possible glueing, which is again the identity function on $\mathbf{C}^\times$, does not. □

2.4. Properties determined at the level of stalks, and sheafification.

Exercise (2.4.A)**.** *Prove that a section of a sheaf of sets is determined by its germs, i.e. the natural map $\mathcal{F}(U) \to \prod_{p\in U}\mathcal{F}_p$ is injective.*

Proof. The natural map sends a section $g \in \mathcal{F}(U)$ to the tuple $(f_p)_{p\in U}$. Suppose $(f_p)_{p\in U} = (g_p)_{p\in U}$ for all p. Then for each p, there exists some open V_p with $f|_{V_p} = g|_{V_p}$. The collection of V_p form an open cover of U, so by the identity axiom we know that $f = g$. □

Exercise (2.4.B)**.** *Prove that any choice of compatible germs for a sheaf of sets $\mathcal{F}$ over U is the image of a section of $\mathcal{F}$ over U under the map in the previous exercise.*

Proof. Let $(s_p)_{p\in U}$ consist of compatible germs, i.e. for each $p \in U$ there is an open set $U_p \ni p$ and some $\tilde{s}_p \in \mathcal{F}(U_p)$ whose germ at $q \in U_p$ is s_q. We know that the $\{U_p\}$ are a open cover of U. Take $p, p' \in U$, and consider $\tilde{s}_p$ and $\tilde{s}_{p'}$. Since we have compatible germs, we know for each $q \in U_p \cap U_{p'}$ that $(\tilde{s}_p)_q = (\tilde{s}_{p'})_q$, i.e. there is an open set $V_q \subset U_p \cap U_{p'}$ containing q where the restrictions $\tilde{s}_p$ and $\tilde{s}_{p'}$ agree. Since this holds for each q, and the V_q form an open cover of $U_p \cap U_{p'}$, we get by the identity axiom that $\tilde{s}_p$ and $\tilde{s}_{p'}$ agree on the intersections. Therefore, by the glueability axiom there is a section $s \in \mathcal{F}(U)$ that restricts to $\tilde{s}_p$ on each U_p. Moreover, the germ of s at p is the germ of $\tilde{s}_p$ at p, which is s_p, completing the proof. □

Exercise (2.4.C)**.** *If ϕ_1 and ϕ_2 are morphisms from a presheaf of sets $\mathcal{F}$ to a sheaf of sets $\mathcal{G}$ that induce the same map on each stalk, show that $\phi_1 = \phi_2$.*

Proof. Following the hint, we consider the following diagram

$$\begin{tikzcd} \mathcal{F}(U) \arrow[r, shift left, "\phi_1(U)"] \arrow[r, shift right, "\phi_2(U)"'] \arrow[d] & \mathcal{G}(U) \arrow[d, hook] \\ \prod_{p\in U}\mathcal{F}_p \arrow[r] & \prod_{p\in U}\mathcal{G}_p \end{tikzcd}$$

But the maps induced by ϕ_1 and ϕ_2 agree on the stalks, so there is just one bottom arrow making the square commute for either $\phi_1(U)$ or $\phi_2(U)$, i.e. for each $f \in \mathcal{F}(U)$ and $p \in U$, we have $(\phi_1(U)(f))_p = (\phi_2(U)(f))_p$. By injectivity (Exercise 2.4.B), we get that $\phi_1(U)(f) = \phi_2(U)(f)$ for all U and f, so $\phi_1 = \phi_2$. □

Exercise (2.4.D)**.** *Show that a morphism $\phi : \mathcal{F} \to \mathcal{G}$ of sheaves of sets is an isomorphism if and only if it induces an isomorphism of all stalks.*

Proof. For the forward direction, recall the definition of the induced map of stalks (Exercise 2.3.A). It is obvious that if ϕ is an isomorphism then each map on stalks ϕ_p also has an inverse given by the inverse of ϕ.

For the backward direction, let $f, g \in \mathcal{F}(U)$ and suppose $\phi(U)(f) = \phi(U)(g)$. It follows that for each $p \in U$,

$$\phi_p(f_p) = (\phi(U)(f))_p = (\phi(U)(g))_p = \phi_p(g_p).$$

Since the stalk maps are all isomorphisms, we have equality of germs $f_p = g_p$ for all p, so by Exercise 2.4.A we know that $f = g$. For surjectivity, choose $g \in \mathcal{G}(U)$. This yields a compatible family of germs in $\prod_{p\in U}\mathcal{G}_p$, and by the isomorphism also a compatible family $(f_p)_{p\in U}\mathcal{F}_p$ with $\phi_p(f_p) = g_p$ for all p. By Exercise 2.4.B, the family $(f_p)_{p\in U}$ comes from a section $f \in \mathcal{F}(U)$. Now using the same square as the previous exercise, we find that $g_p = \phi_p(f_p) = (\phi(U)(f))_p$, so by injectivity on the level of germs $\phi(U)(f) = g$, showing surjectivity. □